Supplementary Material for "Telling Night from a Failing Camera: Context-Gated Test-Time Adaptation for Maritime Detection"

Laksh Saroha
Department of Electronics and Communication Engineering
Thapar Institute of Engineering and Technology
Patiala, India
lsaroha_be24@thapar.edu

Sandeep Mandia
Department of Electronics and Communication Engineering
Thapar Institute of Engineering and Technology
Patiala, India
sandeep.mandia@thapar.edu

This supplement holds Appendices A–J of the paper. Section numbers in Roman numerals, tables numbered I, II and Figs. 1 and 2 refer to the paper; tables and figures numbered S1, S2, ... are in this supplement.

Appendix A: Dataset details

A.1 Verified counts

All counts below were re-derived from disk on 2026-09-10 and supersede earlier estimates in the project record.

Quantity	VIS	IR	Total
Images	127,309	31,010	158,319
Boxes	962,960	220,776	1,183,736
Paired VIS–IR frames			28,388

Per-run image counts (VIS / IR): pohang00 21,768 / 10,918; pohang01 24,473 / 11,995; pohang02 27,795 / 6,175; pohang03 27,085 / 1,922; pohang04 26,188 / 0. Paired rows per run: pohang00 10,786; pohang01 11,990; pohang02 3,739; pohang03 1,873; pohang04 0. Pairing is defined by the dataset's own timestamp table, not by frame ordinal: 16,544 of the 28,388 pairs have different VIS and IR indices, with per-run offsets ranging from -155 to $+1$.

A.2 Preprocessing

Visible frames are letterboxed, not stretched, from 2048×1080 to 640×338 inside a 640×640 canvas (scale 0.3125, 151 px of pad value 114 top and bottom, area interpolation). Stretching would distort VIS (1.9:1 native) and IR (1.25:1) by different factors and hurt cross-modal overlap. Forty-seven percent of every stored VIS frame is therefore pad, a fact that matters in Appendix A.4. Full-resolution originals are archived and hard-linked in a native-resolution twin used for main-backbone training.

Infrared frames are delivered as 8-bit grayscale by per-frame min–max normalization (each frame's own minimum to 0 and maximum to 255) and letterboxed from 640×512 with 64 px of pad top and bottom. This choice has consequences we disclose: thermal crossover can be visually masked because a near-isothermal vessel is stretched to full local contrast, there is no cross-frame radiometric comparability, and the raw 16-bit data cannot be recovered from the product. The 16-bit dynamic range is modest (median span 702 counts, about 7.72 effective bits), but 51 percent of frames are more than 1.5× range-inflated by outlier hot pixels. A percentile-clip re-export was built, tested and rejected: it gained 0.0011 mAP50-95 (1.75 percent relative) but lost 1.6 points of recall.

A.3 Splits

The first split delivered with the tooling interleaved frames (train N, validation N+1), which leaks near-duplicate frames at 10 Hz. It was replaced by contiguous per-run blocks, and then, when the server-side validation set came out 77 percent buoy against a 5 percent global share, by an interleaved K-block split: each run's shared timeline is cut into K equal blocks in a cycle of ten (block index 4 to validation, 9 to test, others to train, giving 80/10/10 by construction), with guard bands at every boundary and K raised until leakage, balance and coverage gates all pass. The same ordinals are used for both modalities so that VIS–IR pairs and stereo pairs never straddle a split. Table S1 gives the resulting sizes.

Per-run VIS (train/val/test): pohang00 16,376/1,672/1,912; pohang01 18,826/2,068/1,629; pohang02 20,424/2,690/2,864; pohang03 20,962/2,579/2,649; pohang04 19,687/2,343/2,391. Per-run IR: pohang00 8,229/836/950; pohang01 8,800/1,034/1,111; pohang02 4,844/247/330; pohang03 1,406/117/127. Both audits pass. Production VIS training uses a stride-2 subset of 48,136 train frames; IR trains on all 23,279.

TABLE S1
SPLIT SIZES (IMAGES)

	Train	Val	Test	Duplicates across splits	Temporal-proximity violations
VIS	96,275	11,352	11,445	0	0
IR	23,279	2,234	2,518	0	0

A.4 The night-box filter and its reversal

This is the dataset event that most shaped the project. On 2026-07-15 a train-only filter removed every box in any pohang01 frame whose content-median luminance was below 100 on a 0–255 scale, on the reasoning that a box in a black frame is a modality-copied annotation rather than something the visible camera can see. The filter emptied 17,502 of 96,275 train label files and dropped 132,688 boxes (126,948 ship, 5,740 buoy). Validation and test were never touched.

The filter had a bug. Its pad-detection constant treated only near-black pixels as padding, but VIS letterboxing pads with the value 114, so the "content median" was dominated by the 47 percent of gray pad pixels. The statistic that actually ran was closer to "frames whose content 95th-percentile luminance is below 100." A per-box audit later scored 120,829 of the deleted boxes on intensity, gradient and contrast tests. Only 38,135 failed all three; 82,694 would not individually have failed but were deleted because their whole frame was cut, and the flagged and kept populations overlap on two of the three axes.

A pre-registered restoration (2026-09-02) put all 132,688 boxes back and re-dropped only the 38,135 individually flagged, a net gain of 94,553 boxes. The endpoint was night-only VIS mAP50-95 with bands DEAD below 0.005, WEAK 0.005–0.02, ALIVE at or above 0.02, and a day guard floor of $-\max(2\cdot\text{sd_paired}, 0.002)$. The label accounting reconciles exactly (Table S2).

TABLE S2
TRAIN-SCOPE LABEL ACCOUNTING

State	Boxes	Train-label hash
Pre-filter	749,579	`fd60c0834fdd`
After frame-level cut	616,891	`287b11c50b5a`
After full restore	749,579	—
After re-dropping 38,135 flagged	711,444	`8ed69b5974ed`

The result is in Appendix D.4. Two further provenance facts belong here. First, a widely quoted post-restore hash `b92739202127` is a tree-scope hash over all 127,309 VIS label files, not the train-scope hash; the train-scope value is `8ed69b5974ed`. Second, on 2026-09-03 at 21:19, 7,591 pohang01 train label files were rewritten back to pre-filter content with no project script running; the cause is unexplained. An append-only hash ledger now records label state so that any recurrence has a bounded window, and it separately found 8,237 orphan label files belonging to no split list.

A.5 Holdout and contamination

pohang04 (26,188 VIS images, no IR) is the held-out run, but its visible labels are **not unseen**, and we state exactly what has and has not touched it. Before the Phase 3 pre-registration, its VIS frames were in the standard training lists: the earlier VIS detectors, and the VIS ensemble and MC-Dropout arms, trained on 9,841 pohang04 frames. 2,343 of its frames were also pooled into an earlier VIS validation list, and two VIS-only probes scored that list on 2026-09-02: day-only re-ranking and oracle re-ranking (Appendix E.4). What never touched pohang04 is the fusion system: no fusion score was ever computed on it, since it has no thermal frames in any list. The ten Phase 3 checkpoints were never trained on it, and the Mahalanobis references built from them were never fitted on it. The held-out claim of Appendix G rests on those facts and no wider one.

Phase 3 lists were built by filtering the existing lists to remove pohang04 rows, preserving every surviving frame's stride identity: VIS train 48,136 to 38,295, val 11,352 to 9,009, test 11,445 to 9,054. IR lists needed no change. Removing an all-day run shifts composition: validation night share rises from 18.2 percent to 23.0 percent, so Phase 3 numbers are not comparable to earlier pooled validation numbers.

A contamination audit found that the Mahalanobis reference list used to fit the earlier OOD scorer contained 819 of 4,000 frames (20.5 percent) from pohang04. For Phase 3 the references were rebuilt from the retrained checkpoints, VIS from a clean list of 3,181 frames and IR from its unchanged 4,000-frame list. All ten reference caches contain zero pohang04 frames (verified per seed). The paired evaluation lists that feed every benchmark cell (2,232 frames) contain no pohang04 frames.

APPENDIX B: CORRUPTION MODELS

B.1 Two generations

The development benchmark uses 2,232 paired VIS–IR frames from the validation split. Eight cells cross four conditions (clean, fog, low-light, glare) with day and night. Later grids add IR-side corruptions for ten or eleven cells. Adverse conditions

are simulated, not field-collected, and two generations of corruption exist. The v1 corruptions (Albumentations 2.0.8; [1]) produced every result recorded before 2026-10-10 and were then found to be broken at the source: v1 fog ends in a fixed 21-px Gaussian blur at every severity, so its cells measured a blur of the 12-px median target; v1 low light is an additive clip that blackens 82 percent of a day frame and 99.9 percent of a night frame; v1 glare lands in the letterbox pad on 45 percent of frames; every v1 filter rewrites the pad; and IR received visible-light filters, including per-channel color noise. The v2 corruptions reproduce the camera and the dataset's processing in physical order (atmosphere, optics, sensor noise, exposure and the IR per-frame stretch, quantization) on the content rows only. Fog is Koschmieder attenuation in linear radiance on per-frame metric depth [2], from Depth Anything V2 [3] fitted to the flat-sea range of the frame's own attitude, with visibility bands from the fog definition and the IR frame re-stretched as the dataset's conversion does. Low light is exposure scaling to the real night level with the night noise measured on training frames. Glare is a source placed against the frame's computed horizon, whose spread, glint and exposure response were fitted to training frames: the one real sun-entry event by day and the real night lights by night. Sensor noise removed by fog or blur is restored. Rain, motion blur and VIS shot noise remain ImageNet-C-style severities [4] without a physical anchor. Every constant is labeled measured, fitted or assumed, and the assumed ones carry sensitivity rows (`docs/eval/corruption_v2/README.md`). Table S4, Figs. S2 and S4, Table S5 and the corrupted rows of Table S13 report v2. Table S3, Tables S7, S8, S9 and S10 and Table S11 are v1 records and are kept as recorded; R-D1 and Stage 1 are re-run under v2 as robustness checks (Appendix E.2, Appendix E.3). Night frames come from a single run, pohang01. Day-only slices on 9,284 frames are used where noted.

B.2 How much the cells rest on assumed constants

Each assumed or one-event constant was moved alone, at one corruption draw, with everything else as shipped (`docs/eval/corruption_v2/sensitivity.md`). The glare source intensity matters most. Halving it raises glare/day VIS by 0.036 and doubling it lowers it by 0.036; at double intensity, glare/night VIS falls from 0.082 to 0.060, below IR's 0.0687, and the night veto there becomes right. So the glare/night failure, at -0.0146, holds for the intensity fitted to the real lights but not for one twice as bright. The exposure response moves glare/day by -0.0069 (no AE) to +0.0035 (a fully converged AE), and fog by less than 0.001. The IR fog extinction ratio, an assumption from the literature, moves the clean-VIS, fogged-IR cell by -0.0086 to +0.0070 by day and -0.0261 to +0.0139 at night over the range 1.0 to 0.3. Revision 2's Lorentzian glare, whose tail lifted the dark floor of every night frame, would have cut glare/night VIS to 0.014 and switched the night veto off on every frame (veto rate 0 against 100 percent). The defect had been found from the gate's statistics before any v2 detection was scored; this row measures what it would have done.

APPENDIX C: STATISTICAL PROTOCOL

C.1 Tune and test discipline

From 2026-09-01 onward every lever was tuned on TUNE = pohang00 (836 paired frames) and reported on TEST = pohang02 + pohang03 (364 frames), with pohang01 excluded from fitting. The split caught a real overfitting trap immediately: a support IoU of 0.55 won on TUNE but lost on all six held-out variants, while 0.30 won on all six. Under the earlier single-set practice both would have read as wins and the wrong one would have been adopted.

We state plainly that this repository contained no untouched test set before the pohang04 look. pohang00 and pohang01 were held out of gate fitting but scored repeatedly; pohang02 and pohang03 were declared TEST after the fact and already fail a model-selection-bias test, because a candidate was rejected specifically for losing on them; and the code's "fit" selector was literally identical to its "day" selector, so every fit-run day constant was reported on the frames it was tuned on. A `role="final"` evaluation context now structurally refuses any frame selector that spans scoring frames, and it is the mechanism that protects the single look in Appendix G.

C.2 Noise floor

Deltas between systems are always paired on the same frames and the same corruption draw. Pairing tightens the standard deviation of a delta relative to unpaired resampling by 3–16$\times$ on the nine informative cells (11–12$\times$ on most; the two zero-information cells have paired sd 0.0000, which is where the quoted 52$\times$ comes from) (`runs/eval/delta_noise_floor.md` Section I). The combined draw-plus-bootstrap two-sigma floor on a paired delta is 0.0014–0.0031 AP on most cells (full range 0.0000–0.0031). The buoy class carries 74–75 percent of macro-metric variance while making up 5.3 percent of day ground-truth boxes, and two of eleven cells carry essentially no buoy-variance information. These floors were measured on the macro over ship and buoy, while the Phase 3, Stage 1 and held-out verdicts are on ship AP, so the floor was re-measured on ship AP with the same arm, cells, draws and resamples (`docs/eval/delta_noise_floor_ship_2026-10-08.md`). The ship floor is 0.0008–0.0024 on the nine informative cells (full range 0.0000–0.0024), and its maximum, the value that sets the magnitude floor in Appendix C.3, is 0.0024 against the macro's 0.0031. The level sd of ship AP is up to twice the macro's (`runs/eval/metric_noise_floor.md` Section II), but that does not carry over to a paired delta: where buoy AP is constant, the macro delta and its noise are both exactly half the ship delta's, and where buoy AP moves, the noisy buoy delta (600 day boxes) raises the macro floor above the ship floor. A delta below the floor is reported as "not resolved," never as "no effect."

C.3 Dependence-aware intervals

Frames at 10 Hz are autocorrelated, and a frame-level iid bootstrap underestimates the standard error. We measured the inflation with moving-block bootstraps [5] up to block length L = 20 (2 s), the longest the shortest run (pohang03, 117 paired frames) permits: intervals widen by 1.9× to 1.99×. Two qualifications apply. First, this is a lower bound, because the block length is capped by the shortest run, not chosen from the autocorrelation. Second, it was measured on deltas between VIS uncertainty arms and then applied to fusion cells, where it was not re-measured. From 2026-09-10 onward every new interval is a block-bootstrap interval, and the factor 1.95 sets the Phase 3 magnitude floor at 0.0060 rather than 0.0031. Built the same way from the ship-AP floor (Appendix C.2) it would be 0.0047, so 0.0060 is conservative for the ship-AP verdicts, and no recorded verdict changes at 0.0047 (Appendix D.3, Appendix E.3, Appendix G). Night, being a single run, has no block length at which a between-night-run interval is estimable.

C.4 AP convention and class set

All absolute AP values use local linear-interpolation AP with the project's maximum detection count, not COCO AP [6]. The two conventions disagree on deltas by at most 0.000285, five times below the noise floor, so no decision can flip on convention; they disagree on absolutes by up to −0.0050, so every absolute value states its convention. The one table outside this convention is Table S16, the backbone benchmark: it reports the training library's own validation mAP (Ultralytics 8.4.90, ship and buoy), because it is read from training logs, says so in its caption, and none of its values is set beside a local-AP number. Two training-library versions (8.4.7 and 8.4.90) disagree on mAP50-95 by about 0.034 for identical weights and data; no table in this paper places numbers from different library versions side by side.

Class set. Ship is the primary class. It is the only class both detectors emit, since the IR detector is single-class (Appendix D.2), and the Phase 3 pre-registration fixed fused ship AP (class 0) as its quantity before any Phase 3 number existed (`docs/prereg-phase3-retrain-2026-09-10.md`). Tables S4, S8, S10 and S13 and Figs. S2 and S3 report ship AP. Table S16 reports the macro over ship and buoy. Table S7 reports ship AP, its registered metric, with the recorded macro beside it as Table S9 (Appendix E.2). Table S6 reports both classes for each stream. Table S3 and Table S11 (Appendix E.4) report ship AP, except two Table S11 rows that stay on the macro and say so: day-only re-ranking, whose substrate includes pohang04 and cannot be re-scored, and VIS soft-NMS, whose registration scored the macro and whose failing cell is at night, where the two are identical because no night frame has a buoy. Table S11 rows recorded on the macro were re-scored on ship from cached detections, each after its unchanged macro path reproduced the record (`docs/eval/class_set_audit_2026-10-08.md`). The constants re-price in Appendix D.3 is reported on the macro, which its registration scored. The macro is not a safe stand-in for ship AP, for three reasons. A VIS veto deletes every buoy, because the surviving IR stream cannot supply one, so a macro delta can move by half a class for a reason unrelated to ship detection. Buoys carry 74–75 percent of the macro's variance on 5.3 percent of the day boxes (Appendix C.2). And a single-class stream scored on the macro reads exactly half its ship AP (Table S6). R-D1 shows the cost: its macro read NULL where its registered ship AP reads POSITIVE (Appendix E.2).

C.5 Metric contracts

Six claims about the uncertainty metrics were tested and hold: D-ECE, here the confidence-only form of the detection calibration error of Küppers et al. [7], conditions on confidence only; AUSE [8] and AURC [9] are ranking-only (a rank-reversing control moves AUSE from 0.0630 to 0.3569); NLL and interval-ECE are computed on true positives only and are published with their true-positive share and recall denominators; AURC is a grid mean, whose gap to the trapezoidal integral (0.0215) is published alongside. One defect was found and is disclosed rather than repaired: WBF can emit fused confidences above 1.0 (maximum 1.7532, on 0.0641 percent of detections) because two overlapping same-stream boxes count as confirmation. This traces to WBF mechanics, not to cross-modal support.

C.6 Pre-registrations and decision rules

The verdicts below were fixed in advance; the other results in Appendices D–F, including Table S3, most of Table S11 and the first σ re-ranker ablation, are descriptive and are labeled so. The registrations and their rules are: the night-label restore (ALIVE bands and a day guard); the re-pricing of inherited constants (margin 2·hypot(sd_draw, sd_paired)); soft-NMS [10] adoption (every cell non-negative, draw-averaged); the R-D1 mechanism ablation (at least three of four conditions above a 0.0060 floor with block-bootstrap CI excluding zero); the Phase 3 retrain with nine append-only amendments; the Stage 1 correspondence crossing (non-inferiority within 0.0060 on at least three of four conditions); the σ re-ranker replication (σ's paired increment at or above 0.0060, or 0.0047 for a weaker pass, with block-bootstrap CI above zero, on at least four of five Phase 3 seeds; Appendix E.4); and the pohang04 single look (Appendix G). After the interval and convention corrections were applied retroactively, 20 of 74 previously significant fusion findings became indeterminate; the large effects (removing the veto on night, fog and glare; the veil repair at +0.0716) survived. That count widens the recorded intervals by the 1.95 factor rather than re-scoring each finding, so it inherits the factor's qualifications from Appendix C.3.

C.7 Identity checks and power

A one-frame shift in cache pairing moves gated fusion by -0.000968, below the noise floor and therefore undetectable by any statistical test; it is catchable only by a content identity check, which the evaluation code now performs on every cache load. The split fingerprint was found to be label-blind (deleting a box leaves it unchanged) and was supplemented with a label fingerprint. Before spending Phase 3 compute we computed minimum detectable effects: with five seeds, VIS resolves 0.01291 and IR 0.01010, neither reaching the 0.0060 floor, so the pre-registration cut the retrained-versus-deployed comparison in advance and it is not reported.

APPENDIX D: THE SHIPPED SYSTEM AND HOW IT BROKE

D.1 As designed

Each modality runs its own YOLO backbone with separate weights. From each backbone a Gaussian σ^2 head produces per-box aleatoric uncertainty and a Mahalanobis distance on backbone features produces a frame-level out-of-distribution score. Per frame and per modality m:

- size-normalized per-box uncertainty $u_i = \frac{1}{4}\sum_t \sigma_{i,t}/s_{i,t}$ over t in {x, y, w, h}, with s the box width or height;
- confidence-weighted frame aggregate $U_{\mathrm{box},m} = \sum_i c_i u_i / \sum_i c_i$;
- calibrated OOD score $O_m = \mathrm{sigmoid}((d_m - \mu_d)/\tau)$;
- reliability $R_m = r_{\mathrm{frame},m} \cdot r_{\mathrm{box},m}$ with $r_{\mathrm{box},m} = \exp(-\lambda U_{\mathrm{box},m})$ and $r_{\mathrm{frame},m} = 1 - O_m$, falling back to $r_{\mathrm{frame},m}$ on empty frames;
- temporal smoothing $\bar{R}_m(t) = \alpha R_m(t) + (1-\alpha)\,\bar{R}_m(t-1)$;
- fusion weight $w_m = \bar{R}_m/(\bar{R}_{\mathrm{vis}} + \bar{R}_{\mathrm{ir}})$ fed to Weighted Boxes Fusion (WBF; [11]).

The design document itself marked every constant and the multiplicative form as choices to be validated empirically. They were, and the validation is the subject of this paper.

D.2 As shipped (preset `crossmodal126m`)

Detectors. VIS uses yolo26m [12] with two classes. IR uses yolo26m with a P2 feature neck and a single class, because IR cannot see buoys: its buoy AP50-95 is 0.0002. Backbone selection is in Appendix H.

Gaussian head. A fresh log-variance branch (`cv4`) is bolted in place onto the live detection head of a loaded model; there is no fork of the training library. Variance is parameterized as log σ^2 over left-top-right-bottom distances in stride units and converted to pixels at inference, riding through post-processing as extra channels. Training adds a fourth loss term with beta-NLL weighting [13] and a warm-up during which the NLL weight is zero. The σ branch reads detached features and the NLL sees a detached mean, so the deterministic detector is intended to train identically to the baseline by construction; this parity is not yet demonstrated. Porting to YOLO26's end-to-end head forced three changes: σ rides the one-to-one branch only, because inference decodes from it; post-processing is overridden to gather σ with the boxes' top-k index; and the NLL target is left unclamped because at reg_max = 1 the stock clamp collapses every target to a constant. One ablation, training σ on undetached features, cannot be run on end-to-end heads because the library detaches the branch upstream.

Mahalanobis OOD score. Lee et al. [14], backbone features are captured by a forward hook and scored against a Ledoit–Wolf covariance [15] fit on a reference set of clean training frames. In the shipped preset the parameters `mu_d = 1e9` and `lam = 0` make this score mathematically inert in the fusion weight. It is retained as a diagnostic and discussed in Appendix D.3 and Section VII.

Decision layer. The shipped decision layer is a hard sensor-selection veto followed by union aggregation (Fig. S1).

1) *Night vote.* The IR stream is asked whether it is night by a 5th-percentile luminance test (`ir_p05 > 41.5`, fitted once on clean IR and frozen). IR may cast that vote only while it still looks like IR: an 11-statistic multivariate health score over the IR frame (gradient Gini, the ratio of Laplacian to variance and nine other frame statistics) must stay within an authority bound at the 99th percentile of clean statistics. `night` is the vote of a healthy IR frame. The fit is not out-of-sample. The threshold is a midpoint that includes the night minimum of the evaluation night run, and the health model was fitted on all clean paired IR frames, night included. Their 0 percent clean error rates are therefore in-sample.
2) *VIS statistics.* Three single-frame statistics, none temporally filtered. `dark` is a 5th-percentile luminance below 10.5. `veil` is a scale-free Gini coefficient of gradient magnitude (`grad_gini`) below 0.483; it was 100 percent correct on fog and 0 percent false-positive elsewhere during development, on the v1 corruptions. That accuracy belonged to v1's fog, a 21-px blur; on physically modeled fog (v2, Appendix B.1) `veil` activates on 9 percent of fogged day frames and 38 percent of fogged night frames, and on every v2 low-light day frame (`docs/eval/corruption_v2/gate_axes.md`). `concentrated` is a ratio of Laplacian to variance above 4.28, the signature of point highlights on an otherwise empty field at night.
3) *Rule.* VIS is vetoed when `night AND (dark OR veil)`, so a veiled VIS frame need not also read dark. When IR votes night but fails its health check, its vote is not believed but may be confirmed: VIS is still vetoed if its own frame shows `concentrated OR (dark AND veil)`. A vetoed stream is removed from the WBF input list rather than

down-weighted, because WBF renormalizes whatever weights it is handed; down-weighting alone left a genuinely blind stream with weight 0.432.

4) *Single survivor.* A frame with one surviving stream returns that stream's detections untouched.
5) *Merge.* Surviving streams are passed to WBF at IoU threshold 0.85 with constant capability-prior weights: each stream's clean mAP50-95 against VIS ground truth on the 1,200 clean day paired frames, with IR's divided by 4 (VIS 0.3233, IR 0.0024 after the division, so w_{vis} = 0.9926; `runs/eval/levers_26m.json`). The prior excludes the night run but not the day frames that every day cell is scored on. At this threshold only 0.05 percent of VIS boxes have an IR partner, so the merge is concatenation; we document the merge as effectively off.
6) *Cross-modal support.* A score multiplier is applied to boxes with a partner at IoU 0.30 (γ = 0.5). It confirms but never moves a coordinate.
7) *IR dedup.* The IR stream is de-duplicated with NMS at 0.70 before merging.

What is off. `sigma_weighted = False` (σ never moves a fused coordinate), `sigma_score_alpha = 0` (σ never moves a fused score), the Mahalanobis weight is inert as above, and VIS soft-NMS is off following a pre-registered rejection (Appendix E.4). The measured consequence is that the fusion weight w_{vis} takes exactly one value across all 2,232 paired evaluation frames and all four corruption conditions. That value is 0.9926 under `crossmodal26m` (8,928 frame-condition pairs) and 0.9930 under the predecessor `crossmodal`.

Two presets, stated once. `crossmodal` (2026-09-01) vetoes VIS on `veil OR (night AND dark)`, so it drops VIS on every veiled frame, day or night: every fog frame on the v1 corruptions, and every low-light day frame on v2. `crossmodal26m`, which ships, vetoes on `night AND (dark OR veil)`, adds a weak night fallback and the cross-modal support term, and keeps both streams on day fog. Every result below names the preset it was measured under.

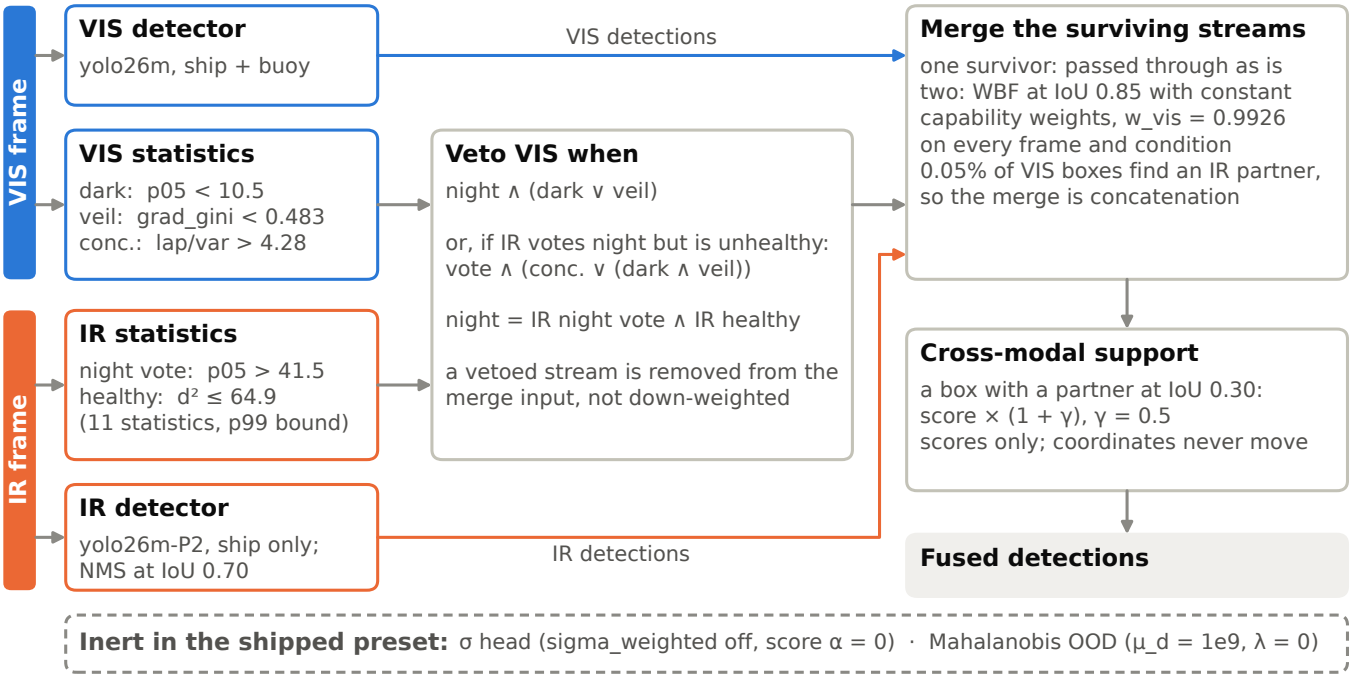


Fig. S1. **The shipped decision layer (preset `crossmodal26m`).** Frame statistics from both streams drive a hard veto on VIS; the surviving streams are merged by WBF with constant capability weights, so w_{vis} = 0.9926 on all 2,232 paired frames and all four conditions, and at IoU 0.85 the merge is concatenation. A cross-modal partner at IoU 0.30 raises a box's score and never moves it. Thresholds are the frozen constants the code loads (`runs/eval/structure_constants.json`, `runs/eval/brightness_constants.json`); the night threshold and the IR health model are fitted in-sample. The σ head and the Mahalanobis score are computed but inert.

D.3 Fusion robustness of the sensor-selection baseline

The decision layer in Appendix D.2 is the sixth rewrite. Table S3 records the worst-cell gap, in ship AP, to the better single stream at each stage; each rewrite was forced by a measured failure of the previous one. **Every number in Table S3 and the paragraph after it was measured on pre-restore checkpoints whose VIS detector scored 0.0000 at night: the yolo26s detectors for stages a–d, and the full-scale yolo26m detectors from stage e on. The paragraph names which. All of it used the v1 corruptions (Appendix B.1) and is kept as recorded.** Table S4 re-measures the shipped rule on the five Phase 3 systems under the v2 corruptions.

TABLE S3
WORST-CELL GAP OF GATED FUSION VERSUS MAX(VIS, IR) ACROSS GATE REWRITES

Stage (date)	Change	Worst cell	Gap
a (08-19)	p05 photometric term, hard veto	lowlight/day	−0.0180
b (08-20)	soft term dropped as redundant; hysteresis	lowlight/day	−0.0180
c (09-01)	Laplacian-variance veil axis	lowlight/day	−0.0180
d (09-01)	crossmodal: IR night vote, grad_gini	—	+0.0000
e (09-01)	detector swap to 26m breaks veil veto	fog/clean	−0.0632
e′ (09-01)	rule reordered to `night AND (dark OR veil)`	—	+0.0000

Ship AP; each gap is against max(VIS, IR) in its own cell. The 08-19 and 08-20 records ranked cells against IR alone, and on that comparison their worst cell was fog/night: −0.0021 [−0.0029, −0.0009] at a, and a tie at b. Against max(VIS, IR), the photometric veto deletes VIS on every synthetic low-light day frame, where VIS still works, and lowlight/day is the worst cell from a through c (`runs/eval/x_fusion_ci.md` Section IV, `runs/eval/final_system.md` Section II, `runs/eval/gate_lab_rules.md`).

Under the shipped preset on the pre-restore full-scale checkpoints, clean/day gated ship AP is 0.3792 against 0.3686 for VIS alone (+0.0106 [+0.0086, +0.0124]), and glare/day is 0.3144 against 0.3024. On the earlier yolo26s checkpoints under `crossmodal`, the same comparison read 0.371–0.374 against 0.368 and 0.296–0.298 against 0.289; the clean/day margin there does not survive the interval correction of Appendix C.3 (`docs/eval/change_impact_2026-09-09_v4.md`). Night cells are bit-identical to IR-only by construction, because VIS is vetoed on 100 percent of night frames and the single survivor is passed through. With a VIS detector that scored 0.0000 at night, that cost nothing. Table S4 shows what it costs once VIS can see. Fog/day, where an earlier veto deleted the better stream, recovers from 0.0192 to 0.0908 after the rule reorder; turning the veil repair off costs −0.0416 on fog/clean, 7.7 times the pre-registered margin (on the macro over ship and buoy, the re-price's registered metric). On the yolo26s checkpoints, removing the veto entirely costs 0.002–0.017 on night, fog and glare, and re-enabling Mahalanobis weighting costs 0.019–0.023 specifically on lowlight/day. Capability-prior weights, the merge threshold of 0.85 and the veil repair were all re-evaluated against the measured noise floor, on that registered macro, and all stand. On the v2 corruptions and the five Phase 3 systems (`docs/eval/corruption_v2/veil_reprice.md`), the veil axis carries nothing with clean IR: removing it changes none of ten cells, because at night `dark` already vetoes wherever `veil` would and by day the shipped rule never asks it. The repair's direction survives. Making the axis unconditional again would veto VIS on every v2 low-light day frame and cost 0.1572 there, plus 0.0071 on fog/day. Under v2 the axis acts only inside the weak-IR fallback of Appendix D.5, which is where that section finds the hole.

Three findings from the rewrite history carry general lessons. First, the original night blindness of the fusion traced to the Mahalanobis reference set, which contained 782 of 4,000 frames from the night run, so night was in-distribution by construction and scored cleaner than day (mean D_night 28.6 versus D_day 30.8). The same raw darkness produced a 21× different fusion response depending on whether it was synthetic low-light (w_{vis} = 0.037) or real night (w_{vis} = 0.792). No corruption ladder could have surfaced this; a day-only refit separates correctly (mean D_night 89.0 versus D_day 30.9). Second, VIS brightness alone cannot distinguish a dark world from a dark sensor: synthetic low-light day has p05 = 0, darker than real night (p05 = 2.5–3.5), yet VIS still works on it. Asking the IR stream whether it is night is the right axis. Third, the veil veto's −0.0632 regression on detector swap shows that a veto encodes a claim about the detector, not about the image: the new VIS detector's own fog AP rose 41× (0.0020 to 0.0824), removing the justification for vetoing it there.

TABLE S4
THE SHIPPED RULE ON THE FIVE PHASE 3 SYSTEMS (VIS SEED K + IR SEED K), CROSSMODAL26M, DEVELOPMENT PAIRED FRAMES (1,200 DAY / 1,032 NIGHT), CLEAN IR, CORRUPTION V2 (APPENDIX B.1), SHIP AP (LOCAL AP)

Cell	VIS only	IR only	Fused (shipped)	Fused − VIS	VIS veto rate	Fused ≥ max(VIS, IR)	v1 record
clean / day	0.3485	0.0218	**0.3566**	+0.0081 [+0.0047, +0.0115]	0%	holds	(no corruption)
fog / day	0.0106	0.0218	**0.0239**	+0.0133 [+0.0065, +0.0200]	0%	holds	0.0601 / 0.0659, holds
lowlight / day	0.1719	0.0218	**0.1790**	+0.0070 [+0.0047, +0.0094]	0%	holds	0.0531 / 0.0607, holds
glare / day	0.1581	0.0218	**0.1701**	+0.0120 [+0.0084, +0.0156]	0%	holds	0.2833 / 0.2940, holds
clean / night	0.2535	0.0687	**0.0687**	−0.1847 [−0.2039, −0.1655]	100%	**fails**	(no corruption)
fog / night	0.0439	0.0687	**0.0687**	+0.0248 [+0.0172, +0.0324]	86.4%	holds	0.0003 / 0.0687, holds
lowlight / night	0.2390	0.0687	**0.0687**	−0.1703 [−0.1862, −0.1543]	100%	**fails**	0.0005 / 0.0687, holds
glare / night	0.0833	0.0687	**0.0687**	−0.0146 [−0.0268, −0.0024]	100%	**fails**	0.1559 / 0.0687, fails
fog s1 / day	0.0270	0.0218	**0.0360**	+0.0090 [+0.0036, +0.0144]	0%	holds	0.0834 / 0.0895, holds
fog s1 / night	0.1066	0.0687	**0.0687**	−0.0378 [−0.0516, −0.0241]	100%	**fails**	0.0004 / 0.0687, holds

AP is the seed mean; corrupted cells average four corruption draws (VIS 941–944) per seed before averaging seeds. Deltas carry the between-seed 95% t-interval (df 4). The last column is the same cell on the v1 corruptions, as recorded on 2026-09-27: VIS only / fused, and the no-loss property.

On the retrained systems, under the v2 corruptions, the no-loss property holds on five of eight cells and fails on three (Fig. S2). The fail criterion was fixed in the exposure ledger before scoring; on the v1 corruptions the same criterion gave six

of eight. The two italic rows are fog at severity 1, the middle of the fog range, added as a descriptive check and not one of the eight cells.

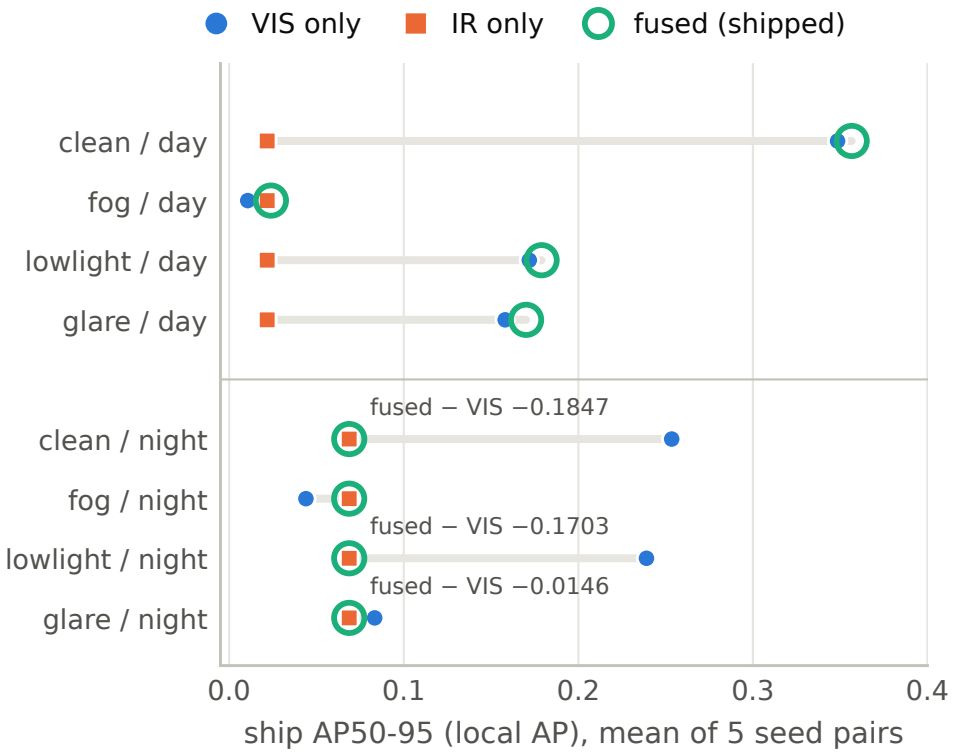


Fig. S2. **The shipped rule on the five Phase 3 systems (Table S4), ship AP (local AP), seed mean over VIS seed k + IR seed k, development paired frames (1,200 day / 1,032 night), clean IR, corruption v2; corrupted cells average four corruption draws (seeds 941–944) per system.** By day the fused output sits just above VIS alone on every cell, and above the clean IR stream where v2 fog takes VIS below it. At night VIS is vetoed on every clean, low-light and glared frame and on 86 percent of fogged ones, so the fused output equals IR alone: right under fog, where VIS falls below IR, and wrong on the other three night cells, where VIS still works. Labeled deltas carry between-seed 95% t-intervals in Table S4.

Day, all four cells. The veto never activates, so the fused output is the union of both streams. It sits above VIS alone on every day cell, with every between-seed interval clear of zero: +0.0070 to +0.0133 (on v1, +0.0059 to +0.0107). The gain is small. v2 fog takes VIS below the clean IR stream by day (0.0106 against 0.0218), and the union still sits at or above both (fused − IR +0.0021 [−0.0022, +0.0064]).

Night: right only where VIS falls below IR. Of the four night cells, only fog at severity 2 takes VIS below IR (0.0439 against 0.0687), and there dropping it is right: the fused output beats VIS alone by 0.0248. On every other night cell VIS still beats IR, and dropping it costs 0.1847 on clean night, 0.1703 under low light, 0.0378 under fog at severity 1 and 0.0146 under glare. v2 low light barely changes a night frame, which is already at the night exposure level, so that cell is close to clean night; v1's additive clip had blacked out 99.9 percent of each night frame, and that is why v1 judged the veto correct there. The rule gates on *darkness*, while what decides whether VIS should be dropped is *VIS health*. With the pre-restore detector, which scored 0.0000 at night, those two variables were the same. After retraining they are not, and under physically modeled corruptions they agree on one night cell of four. The split is the one the earlier night-veto registrations found (Appendix D.4).

For clean/night alone, removing the night arm would give 0.2895, above VIS alone by 0.0360. We do not adopt that. The rule is frozen, and the earlier attempt to remove the arm failed on night cells where VIS was degraded (Appendix D.4).

D.4 Night visible blindness was a label artifact

Every night table before 2026-09-02 showed VIS mAP50-95 of exactly 0.0000. After the restore of Appendix A.4, a 25-epoch fine-tune from the existing VIS checkpoint (early-stopped at epoch 20, best at epoch 10, 5.82 h) gave night-only VIS mAP50-95 of 0.2520 [0.2473, 0.2567] (se 0.0024), 12.6 times the ALIVE threshold (Fig. S3); mAP50 rose from 0.0000 to 0.4957. The day guard passed (+0.0162 against a floor of −0.0045), with the caveat that the day gain is buoy-driven and likely reflects extra training budget rather than the restore.

The visible detector was not blind at night. It was untrained at night. The 0.0000 was manufactured by the filter, and the night arm of the veto had been answering the filter, not the darkness.

The retrained system inherits the rule and pays for it. The five Phase 3 VIS detectors trained on the restored labels and see at night: 0.2535 seed-mean ship AP on the night run. The frozen rule still drops VIS on every clean night frame

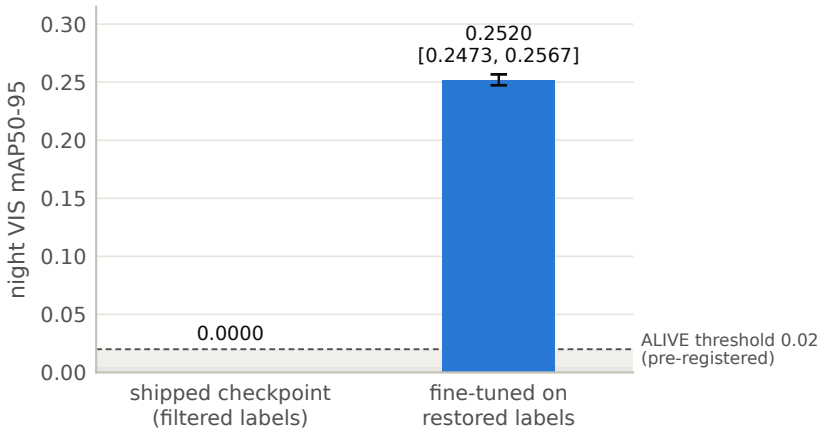


Fig. S3. **Night VIS mAP50-95 before and after the label restore.** 2,068 `pohang01` validation frames, 16,179 ground-truth boxes, all ship, so mAP50-95 is ship AP. The interval is the paired bootstrap interval (1,000 resamples) on the difference, which equals the new arm because the old arm scores 0.0000. The dashed line is the pre-registered ALIVE threshold (0.02); the shaded strips below it are the WEAK (0.005–0.02) and DEAD (< 0.005) bands.

(Fig. S4), so the fused night output equals IR alone, 0.0687, and falls 0.1847 [−0.2039, −0.1655] below VIS alone. Under the v2 corruptions it also falls below VIS by 0.1703 under low light, 0.0378 under fog at severity 1 and 0.0146 under glare; only under fog at severity 2, where the retrained VIS detector does fall below IR at night, is the veto right (Table S4). On the v1 corruptions, glare had cost 0.0872 and the veto had been right under fog and low light, because v1 fog and low light blinded VIS at night. The rule was correct for the detector it was written against and is wrong for the detector it ships with. Nothing in the image changed; what changed is the claim the rule makes about the detector.

Removing the night arm is not the fix. Three pre-registered attempts, run before Phase 3 on an earlier night-trained VIS checkpoint, tried to re-price the arm, and none was adopted.

- V1 removed the arm: +0.1785 on clean night. It was INCONCLUSIVE, because two VIS-degraded night cells fell below IR alone.
- V2 replaced darkness with a VIS health test: INCONCLUSIVE, with five night cells below max(VIS, IR).
- V3 widened the trigger and broke day safety: one cell's day veto rose from 0.8 to 19.9 percent. It was declared VOID.

The measured lesson from that sequence is that the variable which should gate VIS at night is VIS health, not darkness. Under the crippled detector the two coincided. The one health instrument that separated them (AUROC 0.9921) was selected on the only night run, so no held-out night exists to confirm it. The axis was closed before the V3 number was known, and this paper reports the night cost of the frozen rule as a defect of the shipped system, not a tuned repair.

A related repair, adding a VIS-independent confirmation to the veil mechanism, halves the worst night gap from −0.0296 to −0.0141 without touching day.

D.5 Is the IR night switch safe when IR is corrupted?

The night vote trusts the IR stream. IR was uncorrupted in every benchmark cell, so we tested the frozen rule `ir_p05 > 41.5` against IR corruption at three severities, over all 2,232 paired frames. A false night on a clear day vetoes a working VIS stream and is the dangerous direction. Table S5 uses the v2 corruptions (Appendix B.1). The v1 probe also applied IR glare (19.2–26.8 percent false night), low light and rain; those are visible-light filters on thermal frames, and v2 does not model them.

TABLE S5
FALSE-NIGHT RATE OF THE IR NIGHT RULE UNDER IR CORRUPTION, CORRUPTION V2, 1,200 DAY FRAMES

Hazard	Raw, s1	Raw, s2	Raw, s3	Hardened, every severity	Raw separable after a refit
blur	0% [0%]	0% [0%]	0% [0%]	0%	s1 only
fog	13.4% [43.8%]	4.8% [86.7%]	61.4% [94.8%]	0%	no
noise	0% [0%]	16.7% [0%]	96.3% [0%]	0%	s1 and s2

Raw rule `ir_p05 > 41.5`; v1 raw rate in brackets. Hardened = the shipped vote (raw rule and IR health within the authority bound).

The raw rule is unsafe under IR fog and, unlike under v1, under IR noise: after the dataset's per-frame stretch, noise lifts a day frame's 5th percentile toward the night level. The hardened vote calls no false night on any of the nine arms; the health score flags 93–100 percent of fogged and noisy day frames, so a damaged IR stream cannot cause the veto through the main clause. That 0 percent is in-sample: the night threshold and the health model were fitted on data that includes the evaluated night run (Appendix D.2).

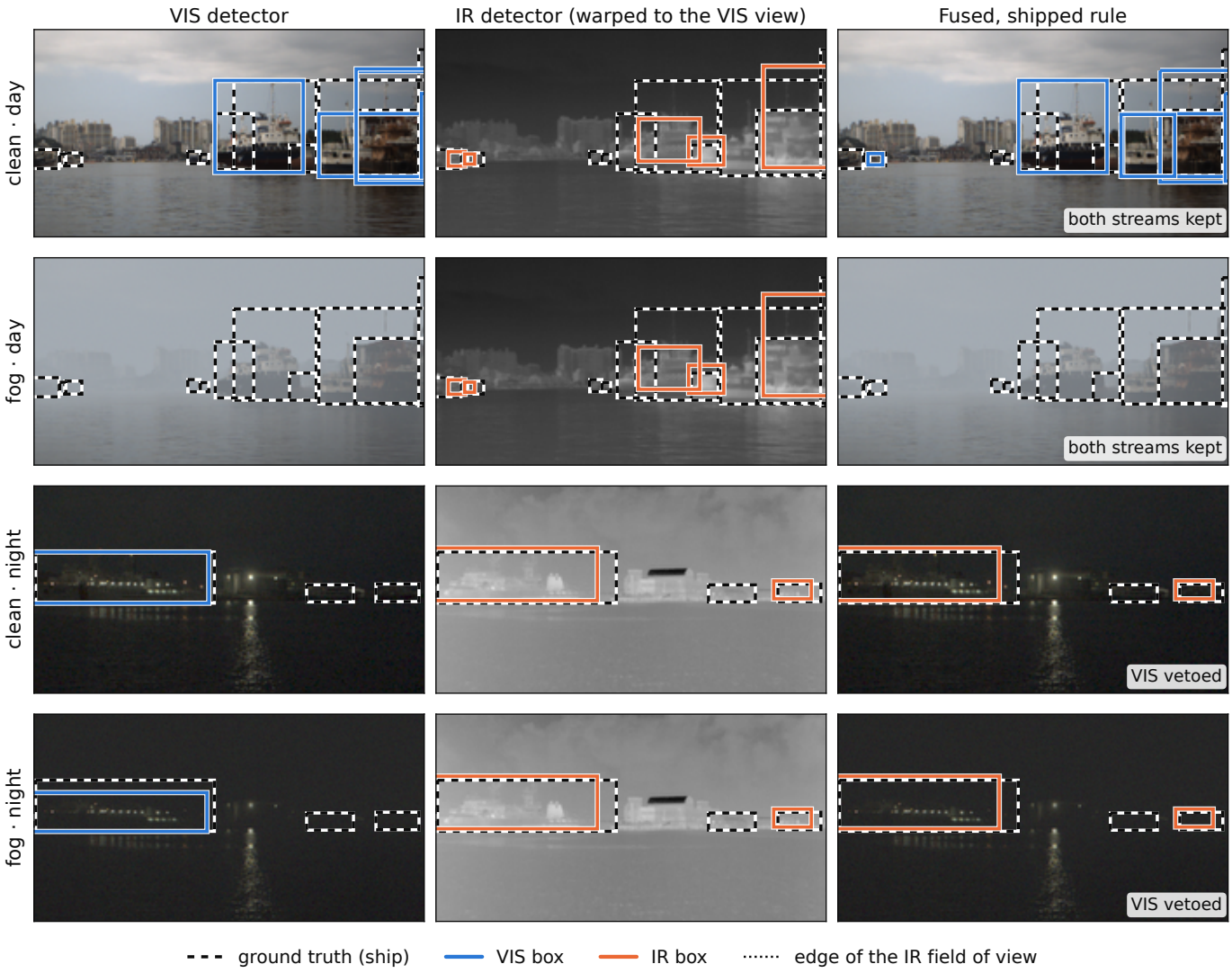


Fig. S4. **The shipped rule on one day frame and one night frame, clean and fogged.** Phase 3 system seed 0 (VIS seed 0 with IR seed 0), preset `crossmodal26m`, development paired frames (pohang03 by day, pohang01 by night). Fog is VIS draw 941 at severity 2 under corruption v2 (Appendix B.1), regenerated with the cache builder's own call; IR is clean in every row, as in every Table S4 cell. Frames were chosen for legibility by visual review of renders that showed ground truth only, never a detection (`scripts/fig_detections.py --candidates`), and the crops were set in the same review. Each panel is a 320 × 170 crop of the 640 × 640 canvas; IR is warped into the VIS view by the per-frame homography the fusion uses; night VIS is brightened for display only (gamma 0.45), and the detector saw the stored frame. Ship class only, boxes at confidence $\geq$ 0.25, fused boxes colored by the stream they came from. By day the veto does not activate and both streams enter the merge, but IR boxes enter with their scores scaled by the IR capability weight (Appendix D.2), below 0.02 here, so the fused output shows VIS boxes. Two VIS boxes below the display threshold (0.20 and 0.18) appear in it because an IR box overlapping each at IoU $\geq$ 0.30 raised its score through the support term (Appendix D.2). Under fog VIS returns no ship box at the display threshold and the fused output shows none either, while the clean IR stream still finds the berthed ships; its boxes enter the merge scaled below the display threshold. At night the IR vote drops VIS on both frames. On the clean frame VIS has found the ferry (best IoU 0.94), and its boxes are discarded anyway; the IR boxes that replace them fit less tightly (0.75 and 0.53), and AP50-95 averages over IoU thresholds up to 0.95. On the fogged frame VIS still finds the ferry (best IoU 0.71) and is dropped as well. One frame shows the mechanism, not the cost; over the clean night cell the cost is the 0.1847 above.

The weak-IR fallback is where v2 finds a hole. Replaying the shipped veto per frame, it never activates on a day frame whose VIS is clean, glared or fogged, under any of the nine IR arms. With low-light VIS it does activate, through the fallback, on 4.8–61.4 percent of day frames when IR is fogged and on 16.7 and 96.3 percent when IR is noisy at severities 2 and 3. The fallback lets a disarmed IR's night call stand when VIS shows concentrated highlights or looks `dark AND veil`, the signature of a fogged night. v2 low light trips both axes, because an underexposed day frame's gradients are dominated by sensor noise; v1 low light, an additive clip, never tripped `veil`. On the v1 corruptions the hardening had taken the both-degraded worst-case false-veto rate from 24 percent to 8.7 percent to 1.3 percent, and left all eight cells bit-identical. On v2 the worst case is 96.3 percent. Priced on the five systems at one corruption draw (`docs/eval/corruption_v2/fallback_cost.md`), the vetoes cost AP on every one of these cells. Low-light VIS scores 0.1712 by day. With IR fogged or noisy at severity 1 or 2, the fused output falls 0.0067 to 0.0125 below it, every interval below zero. At severity 3 it falls to 0.0577 under IR fog (veto rate 62 percent) and to 0.0069 under IR noise (96 percent), losses of 0.1135 and 0.1643. It is the darkness

lesson of Appendix D.3 again: a dark world and a dark sensor look the same to the statistics the fallback reads. A repair suggests itself, because with clean IR the veil axis decides nothing on v2 (Appendix D.3) and acts only here. Priced on the same cells, the fallback without `veil` recovers +0.0074 to +0.0182 at severities 1 and 2, which turns all three failing cells into passes, and +0.0942 and +0.1431 at severity 3, where the cells still fail by about 0.02 because `concentrated` keeps confirming 14 to 23 percent of the false nights. It changes none of the 31 other cells by any amount, including fogged night frames with damaged IR, the case the fallback was built for (`docs/eval/corruption_v2/fallback_fix.md`). We then registered it, with the support repair below, before any number on three fresh corruption draws existed (`docs/prereg-degraded-ir-repairs-2026-10-10.md`). On those draws (VIS 942–944, IR 952–954, all 60 cells, five systems) it harmed no cell at 0.0014, gained $\geq$ 0.0060 on five (+0.0072 to +0.1439, all low-light day with degraded IR) and moved every other cell by exactly zero; the no-loss property holds on 35 of 60 cells against 32 for the shipped rule. It is adopted as the preset `crossmodal26m_r`, a development result; the shipped preset of the pohang04 look is unchanged (`docs/eval/corruption_v2/degraded_ir_repairs.md`).

A degraded IR stream also hurts the union without any veto. With clean VIS and IR fog at severity 3, nothing is vetoed, and the fused output sits 0.0096 [0.0063, 0.0128] below VIS alone. The loss goes through the cross-modal support bonus (Appendix D.2, item 6), which multiplies the score of any VIS box an IR box overlaps at IoU 0.30. With clean IR the bonus is worth +0.0068. With IR fog at severity 2 or 3 it costs 0.0050 and 0.0113, because the scattered boxes of a near-dead IR detector vouch for VIS false positives; without the bonus the cell sits above VIS alone. Dropping an unhealthy IR stream from the merge does not repair this. That switch was measured net-harmful on v1 and is off in the shipped preset, and on v2 it costs 0.0015 to 0.0169 wherever it activates (`docs/eval/corruption_v2/ir_merge_veto.md`). A bonus gated on IR health is the obvious repair, and we registered it beside the fallback fix. It failed: withholding the bonus wherever IR fails its merge-health check harmed eight cells (up to -0.0158, clean VIS with IR noise at severity 2, where the bonus from a noisy but informative IR stream is worth keeping) and gained none. The bonus stays as shipped.

An abstain signal derived from system-level reliability was implemented and demoted to an advisory flag: releasing the veto when both sensors are flagged prevented zero bad vetoes and lost 2,095 correct ones over 76 both-flagged pairs, and under a corrected risk–coverage metric the abstain ordering does not beat random on zero of four conditions.

APPENDIX E: PREDICTED UNCERTAINTY

E.1 Per-modality uncertainty calibration

TABLE S6
PER-STREAM UNCERTAINTY CALIBRATION, THREE ARMS, DAY SLICE (1,200 OF THE 2,232 PAIRED VALIDATION FRAMES), NO FUSION. LOCAL AP. LOWER IS BETTER ON EVERY COLUMN EXCEPT THE TWO AP COLUMNS; BEST ARM PER STREAM AND COLUMN IN BOLD

Stream	Arm	D-ECE	interval-ECE	AUSE	AURC (grid)	NLL (TP-only)	ship AP	buoy AP
VIS	σ head	0.1119	**0.1693**	**0.0731**	**0.4662**	3.91	0.3782	0.2987
VIS	MC-Dropout	0.1316	0.3788	0.1153	0.5709	—	0.3475	0.2731
VIS	ensemble (5)	**0.0840**	0.4888	0.0950	0.5995	—	**0.3877**	**0.3050**
IR	σ head	0.0568	**0.1079**	0.0856	**0.8344**	2.70	0.1414	n/a
IR	MC-Dropout	0.0806	0.4351	0.1296	0.8455	—	0.1382	n/a
IR	ensemble (5)	**0.0477**	0.2576	**0.0775**	0.8983	—	**0.1488**	n/a

Checkpoints trained on restored labels, one per arm: σ head `gauss_vis_seed0_nightfull` / `gauss_ir_seed0_ft`, MC-Dropout `mc_vis_nightfull` / `mc_ir_seed0_ft_refit`, ensemble of five `ens_vis_nightfull_seed0-4` / `ens_ir_seed0_ft`. AUSE and AURC rank boxes by uncertainty, and MC-Dropout and ensemble uncertainties tie wherever members agree (only 57–68 percent of ensemble day boxes have a unique value); across orderings of the ties these two columns move by up to 6×10^{-4} (IR ensemble AUSE 0.0769–0.0777, VIS ensemble 0.0948–0.0951), which changes no ordering in the table. The σ head has no ties. AP is local AP50-95 per class. The IR detector is single-class (Appendix D.2) and has no buoy AP; a macro over both classes reads exactly half of IR's ship AP, because the 596 buoy boxes in the IR day labels score zero. The arm ordering is the same on either.

Why day-only is primary. The substrate is 46.2 percent night from a single run. The pre-registration measured whether night reorders the arms: it compared each metric's between-arm separation on day frames with how unevenly night shifts the arms (ratio r). On VIS, D-ECE, AUSE and AURC fall in the SUSPECT band (r = 0.30–0.60) and interval-ECE is CLEAN (r = 0.05). On IR, AURC is SUSPECT and the rest are CLEAN. Under the registered rule a SUSPECT verdict makes day-only the primary basis, with the pooled and night rows retained as secondary (in the source file). IR is the negative control, since its labels were never filtered, and it lands in the same band as VIS.

One registered rule was amended after it activated. The rule flagging any day-versus-pooled order change as CONTAMINATED activated on a swap inside the noise, and it was then floored to require separation beyond the floor in both orderings. Both verdicts are reported. **As registered, IR reads CONTAMINATED; as amended, SUSPECT.**

What the table says.

- **Ensemble:** the best confidence calibration (D-ECE) and the highest mAP on both streams.
- σ **head:** the best interval calibration by 2.2–4.0$\times$, and the best AURC on both streams. It is the only arm with a likelihood at all, so the only arm whose uncertainty is a usable number rather than a ranking.

- **MC-Dropout:** last or near-last on nearly every column.

No arm wins everywhere, and the ordering depends on whether one asks for calibrated confidence or calibrated intervals (Fig. S5). That is a calibration result, not a detection result. mAP differences here come from one checkpoint per arm and are not a ranking (next paragraph).

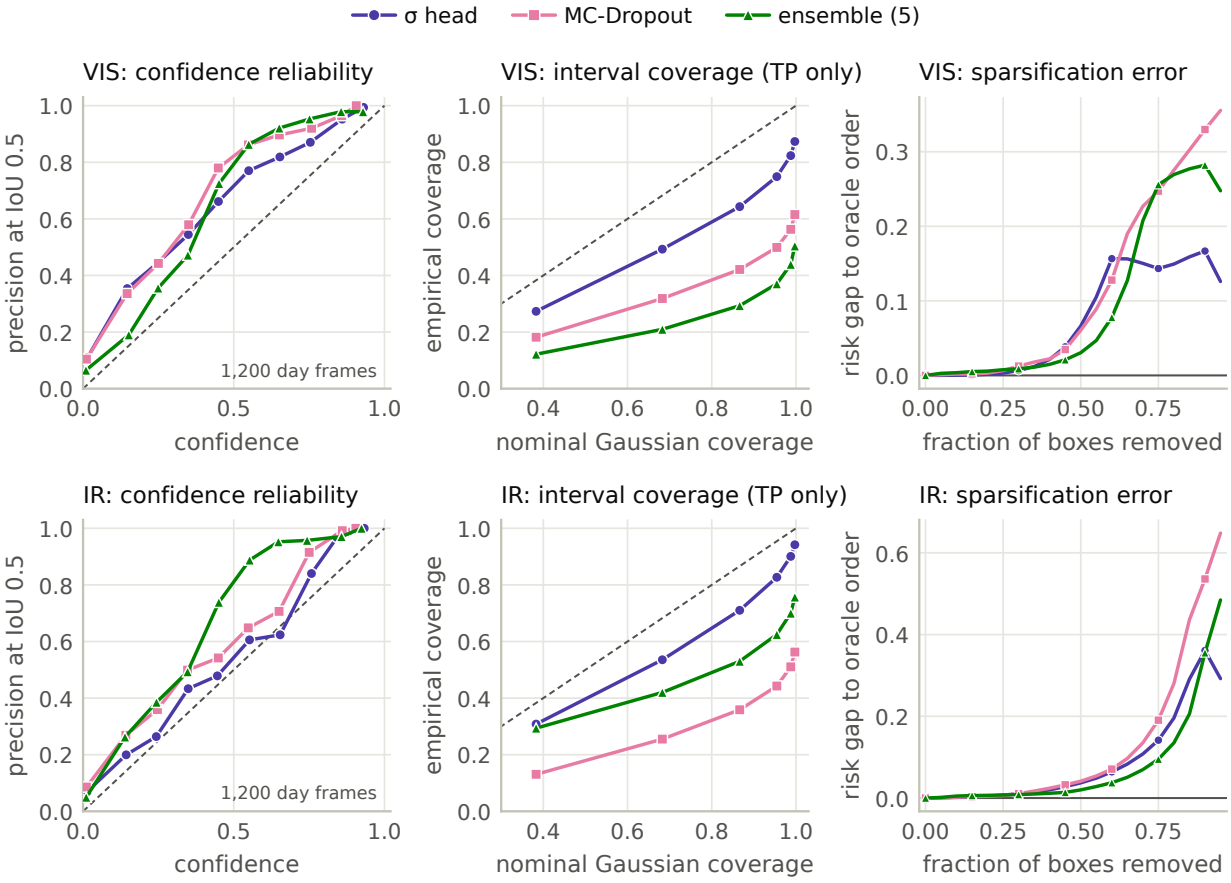


Fig. S5. **Per-stream calibration of the three uncertainty arms, day slice (1,200 of the 2,232 paired validation frames), no fusion; the primary basis of Table S6.** Left: precision at IoU 0.5 against confidence in D-ECE's ten bins (bins with at least 20 boxes). Middle: empirical against nominal Gaussian coverage of $|\text{error}| \leq k\sigma$ for k = 0.5–3, true positives only; for MC-Dropout and the ensemble σ is member disagreement (Appendix E.1). Right: risk (1 − IoU, a false positive counting 1) of the retained boxes minus the oracle ordering's, as the most uncertain boxes are removed; its mean over the grid is AUSE. Dashed lines are perfect calibration. Recomputed from Table S6's caches with the same metric code: D-ECE and interval-ECE reproduce Table S6 exactly, AUSE and AURC to within the spread across tie orders (see Table S6's note).

Scope and caveats.

- These are development-data numbers only. The VIS MC-Dropout and ensemble arms trained on 9,841 pohang04 frames (Appendix A.5), so no three-arm number may appear on the held-out run.
- None of these checkpoints is a Phase 3 checkpoint, and none of the three arms was ever used as a fusion input. R-D1 (Appendix E.2) tests the σ head only.
- The pre-registered estimand for MC-Dropout and the ensemble is **disagreement ranking**, not predictive likelihood. The disagreement statistic omits the within-member variance term and produces $\sigma = 0$ when members agree, which is why their NLL is undefined and the NLL column carries no arm ranking.
- The ensemble's five members form one replicate, not five.
- Night is SUSPECT on both streams. The same band on the negative-control stream means the night miscalibration is not label-driven, at least for these restored-label checkpoints.

The σ head is evaluated here as a localization-error-scale estimate on its own terms, not as a fusion input. Signal screens support that reading: boxes with σ below the per-frame median are true positives 3.00× more often than chance (3.39× at IoU 0.75), the second-strongest signal after confidence (4.80×; `runs/eval/signal_lift_26m.md`). But σ is informative about error magnitude, not direction: leave-one-run-out ridge regression finds live out-of-fold R^2 on only two of four box edges (0.017–0.098), and applying a σ-driven coordinate correction end to end costs 0.02–0.11 mAP.

We do not rank uncertainty methods on mAP. On the best-epoch checkpoint convention an MC-Dropout arm led the ensemble by +4.53 sd; on the epoch-mean convention the same run was the worst arm at −3.00 sd (the two same-machine curves are

in docs/D31-checkpoint-selection-2026-09-01.md). The best checkpoint is a maximum over roughly ten noisy validation epochs and rewards the noisiest run; the between-seed sd of best fitness (0.00212) is 1.7 to 5 times smaller than the within-run epoch-to-epoch sd (0.0035–0.0106). The same recipe run on two machines gave opposite winners depending on the estimator. This is why Table S6 ranks the arms on calibration and not on mAP.

E.2 The pre-registered mechanism test (R-D1)

The system was designed so that predicted uncertainty would weight the fusion. As shipped it does not, and we asked whether it could. The pre-registered comparison is not real σ against a constant, since inverse-variance weighting with equal σ reduces analytically to plain WBF. It is real σ against the same σ values shuffled onto the wrong boxes. Eight arms were scored:

- S0, the system as shipped;
- S1–S4 on the coordinate path: real, constant, shuffled within frame, shuffled across cache;
- S5–S7 on the score path, which multiplies each box's score by (frame-median σ / σ)$^{\alpha}$, with the median taken within the frame and within the box's own stream: real, constant, shuffled within frame, with α fixed at 1.0 before the run and never tuned.

The rule, applied to each path separately, was at least three of four conditions with a delta of at least 0.0060 and a block-bootstrap CI excluding zero. Both pre-registrations name the metric as gated-fusion ship AP. R-D1 ran first under the predecessor preset crossmodal. It was re-run under the shipped crossmodal26m with everything else held fixed, pre-registered before the run. Both use the pre-restore full-scale checkpoints.

A deviation, found late. Both recorded runs scored the macro over ship and buoy, not ship AP. We found this on 2026-10-08, after earlier drafts had reported the macro's verdict, NULL on both paths, as this paper's headline. We then re-scored both presets on ship AP with nothing else changed (docs/eval/uq_mechanism_ablation_ship_2026-10-08.md). The unchanged macro path was re-run first and reproduced both recorded tables exactly: all 32 arm APs and all 24 delta cells per preset. Table S7 is the registered metric. Table S9 keeps the recorded macro as a secondary result.

TABLE S7
R-D1 ON THE REGISTERED METRIC: REAL MINUS SHUFFLED σ, GATED-FUSION SHIP AP, 2,232 PAIRED FRAMES, BLOCK-BOOTSTRAP 95% CI (L = 20)

Path	Condition	crossmodal	crossmodal26m
coordinate (S1 − S3)	clean	−0.000103 [−0.000428, +0.000036]	−0.000126 [−0.000285, +0.000052]
	fog	+0.000000 [0, 0]	−0.000018 [−0.000078, +0.000179]
	lowlight	−0.000003 [−0.000009, +0.000017]	−0.000002 [−0.000009, +0.000019]
	glare	−0.000003 [−0.000139, +0.000097]	−0.000020 [−0.000139, +0.000097]
score (S5 − S7)	clean	**+0.006875** [+0.002661, +0.011151]	**+0.007252** [+0.003290, +0.012387]
	fog	**+0.007919** [+0.005383, +0.010365]	**+0.011957** [+0.009406, +0.014171]
	lowlight	+0.003990 [+0.001887, +0.005368]	+0.003444 [+0.001185, +0.004794]
	glare	**+0.006407** [+0.002266, +0.009536]	**+0.006825** [+0.003113, +0.011398]
passing at 0.0060	coordinate / score	**0 / 3 of 4**	**0 / 3 of 4**

The coordinate path is NULL under both presets, at every floor tested. Real σ against the shipped system with no σ at all (S1 − S0) differs by at most 0.0001 on any condition. Used as the fusion weight it was designed to be, σ changes essentially nothing.

The score path is POSITIVE under both presets. Real σ beats shuffled σ on clean, fog and glare, which meets the bar of three; lowlight falls short (+0.0040 and +0.0034). The verdict depends on the floor, and the registration required counts at several floors so that this would be visible. The score path passes on four of four conditions at 0.0014 and 0.0031, and on three at 0.0047 (the ship-AP floor built as in Appendix C.3) and at 0.0060. At 0.0100 it passes on none under crossmodal and on one under crossmodal26m, which is NULL. Glare clears 0.0060 by only 0.0004 and 0.0008.

On the v2 corruptions the score-path pass does not hold. R-D1 was registered and run on the v1 corruptions, and its verdict is the v1 one. Re-run with only the corruption changed (v2, Appendix B.1; logged in the exposure ledger before the run; runs/eval/rd1_ship_v2_crossmodal_2026-10-10.md, runs/eval/rd1_ship_v2_26m_2026-10-10.md), the coordinate path is NULL again under both presets. The score path passes at 0.0060 on two of four conditions under crossmodal (clean +0.0069, low light +0.0081; fog +0.0042, glare +0.0037) and on one under crossmodal26m (clean +0.0073; fog +0.0041, low light +0.0012, glare +0.0029), so it reads NULL under both. Clean is untouched by the corruption and equals Table S7. Every delta is still positive, with an interval excluding zero on four of four and three of four conditions: real σ still orders boxes better than shuffled σ, but on corrupted frames by less than the floor. The registered POSITIVE therefore depends on the corruption model, and we report it as such. The reading drawn below, informative but not useful, does not depend on it: against no σ (S5 − S0) on v2, the re-ranked system scores lower on five of eight cells (clean −0.0199 and −0.0329, glare −0.0062 and −0.0157, low light −0.0153 under crossmodal26m) and above on three, each by at most 0.0032 (fog +0.0021 and +0.0016; low light +0.0032 under crossmodal, where veil vetoes VIS on every v2

low-light frame, so that cell is IR re-ranking its own boxes), every interval excluding zero (`runs/eval/rd1_ship_vs_s0_v2_{crossmodal,26m}_2026-10-10.json`).

Informative, not useful. Read literally, the registered consequence of POSITIVE is that "predicted uncertainty improves the fusion" for the mechanism that passed. We do not draw that conclusion, because the shuffled control cannot support it. It shows that real σ carries information about which boxes are right; it does not show that using σ beats leaving it out. That second question needs S0, which the arm table contains (Table S8).

TABLE S8
THE SCORE PATH AGAINST NO σ: S5 − S0, GATED-FUSION SHIP AP, BLOCK-BOOTSTRAP 95% CI (L = 20)

Preset	clean	fog	lowlight	glare
`crossmodal`	−0.0199 [−0.0275, −0.0126]	+0.0032 [+0.0012, +0.0046]	−0.0042 [−0.0051, +0.0014]	−0.0150 [−0.0214, −0.0105]
`crossmodal26m`	−0.0329 [−0.0411, −0.0234]	−0.0029 [−0.0056, −0.0010]	−0.0083 [−0.0116, −0.0050]	−0.0266 [−0.0339, −0.0190]

Real σ in the score is below no σ, with a CI excluding zero, on six of eight cells. Under `crossmodal`, lowlight spans zero, and fog scores above S0 only because VIS is vetoed on every fog frame there, so that cell is IR re-ranking its own boxes. Constant and shuffled σ are below S0 on all eight cells. At α = 1.0, multiplying scores by σ costs ship AP, and real σ costs less than shuffled σ; that difference is what passed. The registered response to POSITIVE is to re-examine the shipped preset for adopting the mechanism. Against S0 the re-examination says not to adopt it at α = 1.0, and the shipped preset is unchanged. Whether another α would beat S0 is untested. The registration fixed α in advance because tuning it on the cells being reported would select on them, and no held-out data remains on which to tune it.

The gain is re-ranking within a stream, not fusion. We decomposed the score-path deltas under both presets by emptying the IR stream, descriptively and after the verdict (`scripts/diag_rd1_fog_score_path.py`; `docs/eval/uq_mechanism_ablation_ship_2026-10-08.md` Section IV).

- By day, VIS re-ranking alone, with IR detections removed, gives +0.0160 on clean and +0.0153 on glare under both presets, and +0.0167 on fog under `crossmodal26m`. Adding the IR stream moves these by −0.0040 to +0.0035, with no consistent sign: −0.0008 and −0.0010 on clean and glare under `crossmodal`; +0.0035, −0.0040 and +0.0019 on clean, fog and glare under `crossmodal26m`.
- Under `crossmodal` VIS is vetoed on every fog frame, so fog's day part (+0.0063) is IR re-ranking IR boxes.
- At night VIS is vetoed on every frame, so the night part (+0.0083 to +0.0104) is IR re-ranking IR boxes, on the same uncorrupted frames in every condition.
- Lowlight fails by day. There, real σ does not re-rank VIS boxes (+0.0006, CI spans zero), and lowlight's pooled delta (+0.0040, +0.0034) is the night IR part.

The signal σ carries is therefore a within-detector one: it orders a detector's own boxes better than chance. It does not tell the system which stream to believe, which is the job it was built for.

Why the macro hid it. Buoys are detected only by VIS (Appendix D.2). On fog and lowlight, buoy AP is zero or near it in every arm, so the macro delta is half the ship delta or close to it (`crossmodal` fog: +0.0079 on ship, +0.0040 on the macro). On clean and glare, real σ re-ranks buoys worse than shuffled σ, by 0.0157 and 0.0054 under both presets, which turns the clean macro delta negative and leaves glare's at +0.0005 and +0.0007. The macro thus averaged a halved ship gain with a buoy loss, and read NULL.

TABLE S9
R-D1 AS RECORDED: REAL MINUS SHUFFLED σ, GATED-FUSION MAP50-95, THE MACRO OVER SHIP AND BUOY (APPENDIX C.4). SECONDARY

Path	Condition	`crossmodal`	`crossmodal26m`
coordinate (S1 − S3)	clean	−0.000566 [−0.000893, −0.000041]	−0.000578 [−0.000879, −0.000036]
	fog	+0.000000 [0, 0]	−0.000009 [−0.000039, +0.000090]
	lowlight	−0.000001 [−0.000004, +0.000008]	−0.000001 [−0.000005, +0.000010]
	glare	+0.000406 [−0.000123, +0.000764]	+0.000397 [−0.000124, +0.000761]
score (S5 − S7)	clean	−0.004420 [−0.011030, +0.005829]	−0.004232 [−0.010648, +0.005676]
	fog	+0.003960 [+0.002691, +0.005183]	**+0.006156** [+0.003934, +0.007107]
	lowlight	+0.001995 [+0.000944, +0.002684]	+0.001722 [+0.000592, +0.002397]
	glare	+0.000514 [−0.004569, +0.004948]	+0.000724 [−0.004205, +0.005273]
passing at 0.0060	coordinate / score	**0 / 0 of 4**	**0 / 1 of 4**

Two macro results that earlier drafts reported as pointing the wrong way do not survive on ship AP. On clean, shuffled σ beat real σ on the coordinate path (−0.000566); on ship AP the CI spans zero. On clean, real-σ score re-ranking was worse than a constant (S5 − S6 = −0.013157 [−0.022630, −0.000678] under `crossmodal`, −0.014141 [−0.023552, −0.001021] under `crossmodal26m`); on ship AP, S5 − S6 is positive on all eight cells, so that result was buoy re-ranking. One flaw in

the original rule is self-identified. Under `crossmodal` the coordinate-path bar should have been three of three informative conditions, since fog there is structurally incapable of a coordinate-path effect. The flaw biases against a positive verdict, and the `crossmodal26m` re-run, in which day fog keeps both streams, removes it and returns the same coordinate-path verdict.

Independent probes agree that σ does not improve the fused result. Inverse-variance WBF differs from stock WBF by at most 0.0005 on any cell. Adding σ to the fusion score moves TEST by +0.0010 with a CI spanning zero, and costs 0.0101 on TUNE, where its strength was chosen. A learned VIS re-ranker does gain on ship AP out of fold (+0.0063, Table S11), and σ carries most of it: without σ the same arm gives +0.0014 [−0.0020, +0.0044], and σ's paired increment is +0.0050 [+0.0038, +0.0064]. It orders one detector's boxes, so it is not fusion. It was not registered, and its pre-registered replication on the five Phase 3 detectors FAILS: positive on all five, beyond the floor on too few (Appendix E.4). The Mahalanobis weight was already inert in the shipped preset.

E.3 Relaxing correspondence does not rescue the coordinate path (Stage 1)

The coordinate-path null of R-D1 was measured at a merge threshold (0.85) where almost nothing merges. Two prior negative results on correspondence (a per-frame registration refinement that raised the partner rate 80× to 4.02 percent but lowered held-out AP by 0.0025; relaxing the threshold to 0.55, which cost −0.0138) were each measured with σ inert. Phase 3 Stage 1 crossed the two: threshold {0.85, 0.55} × σ-weighting {off, live}. Cells A (shipped), B (shipped threshold, σ on; bit-identical to A) and C (relaxed, σ off; C − A = −0.0138) were known. Cell D (relaxed and σ live) was the experiment. D had to be non-inferior to A within 0.0060 on at least three of four conditions on TUNE.

TABLE S10
STAGE 1 CROSSING, A − D ON TUNE (POHANG00), BLOCK BOOTSTRAP

Condition	A − D	95% CI	Non-inferior
clean	+0.0151	[+0.0095, +0.0210]	no
fog	+0.0160	[+0.0104, +0.0221]	no
lowlight	+0.0014	[−0.0001, +0.0040]	yes
glare	+0.0115	[+0.0070, +0.0157]	no

One of four: verdict S1-NULL, robust at every floor from 0.0014 to 0.0100. The lone pass is not a rescue; low-light passes because relaxing correspondence barely costs anything there (C − A = −0.0013), not because live σ recovered anything. σ was confirmed live in the plumbing (it changed the fused output on 753–836 of 836 clean frames), yet the interaction terms B − A and D − C sit inside [−0.0002, +0.0004] everywhere with every CI spanning zero. TEST numbers were computed and are reported for completeness: two of four non-inferior, with fog favoring the rejected relaxed setting by +0.0028. As pre-declared, they were not used to override the TUNE verdict. The correspondence question is closed; no threshold below 0.55 will be tried under this project, and the fusion is documented as union aggregation, not consensus.

On the v2 corruptions the count rises to three of four, for the wrong reason. Re-run with only the corrupted VIS caches changed (logged in the exposure ledger before the run; `runs/eval/stage1_crossing_v2_2026-10-10.md`), the clean cell reproduces exactly and still fails, while D is non-inferior to A on fog (+0.0001 [−0.0007, +0.0007]), low light (+0.0032 [+0.0016, +0.0053]) and glare (+0.0002 [−0.0013, +0.0018]). Relaxing correspondence costs almost nothing on those frames (C − A between −0.0031 and 0.0000 on TUNE, against −0.0148 on clean), so D matches A whatever σ does, and the interaction (D − C) − (B − A) stays within ±0.0005 with every interval spanning zero. That is the mechanism of v1's lone low-light pass, now on three cells. The registered verdict is the v1 one, and its substance holds on v2: σ arbitrating relaxed correspondences recovers nothing anywhere.

E.4 Levers that are inert or negative

The one fusion lever with a positive held-out delta is the cross-modal support multiplier at IoU 0.30 (TEST +0.0033 [+0.0010, +0.0067]; a frame-level interval, which spans zero once widened by the factor of Appendix C.3). Merging does not pay across modalities, within a stream or across augmentation views; the test-time-augmentation case is the cleanest, because its views are pixel-exact registered and merging still barely moves. Those rows are screens of agreement as a signal, mostly on the earlier checkpoints. Merged as ensembles by plain WBF at IoU 0.55 on the Phase 3 detectors, flipped views do pay (+0.0075 VIS ship AP), and five checkpoints pay more (+0.0192; Appendix F.2). The one exception on ship AP is merging two VIS checkpoints at IoU 0.55, +0.0069 [+0.0011, +0.0131], the best of six arms; the macro read it as a loss, because the merge costs buoys. The value of a redundancy axis is its independence, not its abundance: confidence lifts true-positive rate 4.80×, σ 3.00×, cross-modal agreement at IoU 0.30 2.08×, and temporal persistence 1.00×, because persistent false positives are the most stable objects in a fixed scene.

The headroom lies elsewhere. Oracle re-ranking of the VIS detections on the 1,200 paired day frames would raise ship AP from 0.3686 to 0.4653 (+0.0968), and on 9,284 day-only VIS frames from 0.3512 to 0.4624 (+0.1112; that substrate includes

TABLE S11
FUSION AND POST-PROCESSING LEVERS TESTED AND NOT ADOPTED. PAIRED DELTAS IN SHIP AP (LOCAL AP), ON THE FRAMES EACH ROW NAMES

Lever	Result	Verdict
σ-weighted WBF †	$\leq$ 0.0005 on every cell	inert
σ in fusion score (α 0.1)	TEST +0.0010, CI spans zero; TUNE -0.0101 [-0.0114, -0.0086]	hurts on TUNE, unresolved on TEST
Per-frame registration alignment, then merge	TEST -0.0010, spans zero (TUNE +0.0121)	does not generalize
Isotonic score calibration	TUNE/TEST -0.0044 / -0.0034	hurts
VIS score re-ranking, leave-one-run-out, 1,200 day frames; inputs confidence, two transforms of it, and σ †	+0.0063 [+0.0030, +0.0094] out of fold, 3/3 held-out runs (in-sample +0.0219); without σ +0.0014 [-0.0020, +0.0044], 1/3 runs; σ's paired increment +0.0050 [+0.0038, +0.0064]	clears 0.0060 by 0.0003 on one checkpoint; σ's increment FAILS its pre-registered replication (2/5 seeds at 0.0060, 3/5 at 0.0047)
VIS score re-ranking, day-only 9,284 frames ‡	best λ = 0, delta 0.0000	null on the macro; includes pohang04, so not re-scorable on ship
Temporal support †	lift 1.00×	dead
Test-time-augmentation view merging †	≈1.0× lift once confidence-matched	inert
Two-checkpoint VIS ensembling †	WBF merge +0.0009 to +0.0069 (IoU 0.55: +0.0069 [+0.0011, +0.0131], best of six arms); concatenation -0.018 to -0.102	merging does not hurt; concatenation does
Within-stream WBF dedup †	-0.0030 to -0.0068	hurts at merge IoU $\geq$ 0.80; unresolved below
VIS soft-NMS σ = 0.5 ‡	night worst cell -0.0000, negative on 4/4 draws	rejected by pre-registered every-cell rule
Capability-ratio alternatives (×4, ×16, ×64)	win clean, lose $\geq$ 1 cell	rejected by no-cell-may-lose rule
Two-sided veto (also veto IR)	worst gap -0.0810 on every night cell, IR uncorrupted	unsafe
Per-stream top-k truncation †	k $\geq$ 100: $\leq$ 0.0001; k = 50: up to -0.0006 (IR, lowlight/day); point deltas	irrelevant

Rows marked † were recorded on the macro over ship and buoy and are re-scored here on ship (`docs/eval/class_set_audit_2026-10-08.md`); their intervals are moving-block, L = 20. Rows marked ‡ stay on the macro.

pohang04 and is read from its existing record); the learned out-of-fold re-ranker of Table S11 recovers +0.0063 of the first on that checkpoint (+0.0025 to +0.0101 on the five Phase 3 detectors). Small objects carry 89.2 percent of ship ground truth with the lowest ship AP; cross-modal union adds only +0.01–0.02 recall over VIS alone. Resolution and ranking, not fusion, are the levers. Resolution is out of scope for a detector held fixed; ranking is not, but the learned re-ranker recovers little of the headroom, and σ's share of it fails its pre-registered replication (below).

The σ increment does not replicate at the floor. Before any of their re-ranker numbers existed, we pre-registered a replication of σ's re-ranker increment on the five Phase 3 VIS detectors (`docs/prereg-sigma-rerank-replication-2026-10-09.md`). It used the same 4-versus-3-feature arm at λ 0.30, ship AP, out of fold on the 1,200 day frames, with block intervals. The outcomes were REPLICATES if at least four of five seeds clear 0.0060 with an interval above zero, REPLICATES AT SHIP FLOOR if four clear 0.0047, and FAILS otherwise. The pipeline first reproduced the original +0.004960 exactly. The increment is +0.0045, +0.0052, +0.0065, +0.0038 and +0.0062 on seeds 0–4, every interval above zero, mean +0.0053. Two seeds clear 0.0060 and three clear 0.0047, so the outcome is FAILS. σ adds a consistent but small within-detector ranking gain, not one this paper can call beyond its floor. The full re-ranker against no re-ranking has an interval above zero on only two of the five (`docs/eval/sigma_rerank_replication_2026-10-09.md`).

The soft-NMS rejection deserves one more sentence. The first pass had no magnitude floor, so a night regression of -1.03e-5 failed the bar exactly as a regression of 1e-2 would have. The project declined to invent an equivalence margin after seeing that it would flip the verdict, and instead measured the noise floor of Appendix C.2. The draw-averaged re-test still failed on four of four draws, and the rejection stands.

APPENDIX F: REPLACING THE GATE

F.1 Replacing the hand-written gate

The shipped decision layer is a hand-written gate that has been rewritten six times, each time forced by a measured failure (Appendix D.3). Under the v2 corruptions it holds the no-loss property fused $\geq$ max(VIS, IR) on 32 of the 60 cells of the development grid (five VIS conditions × six IR conditions × day and night). We registered replacements for it in three rounds, each before any of its numbers existed. All are development results on the Phase 3 systems, scored on corruption draws no earlier system had been scored on, and none touches pohang04. The night axis of `crossmodal26m` stays closed; every replacement is a separate system.

Learned late fusion (LLF). One learned score per box from either stream, in the spirit of CLOCs [16]: 34 box, cross-stream and frame-health features, gradient-boosted trees trained on draw 941 against a matching-credit target, scored leave-one-run-out so that the night run is scored by a model that never saw night (`docs/prereg-llf-2026-10-10.md`). On draws 942–944 it held the no-loss property on 33 of 60 cells (shipped 32) and was worse than shipped on 17 cells beyond 0.0060. It removed the night cost (+0.16 to +0.23 on every clean-VIS night cell) and lost 0.0174 below VIS alone on clean day.

The loss survives removing the frame-health features (claim 31/60) and the cross-stream features (30/60): what both ablations keep is the box-level features, which let the model reorder a stream's own boxes, and a model fitted to a noisy credit target reorders them worse than the detector's confidence does. Both registered predictions FAIL (`docs/eval/llf/llf_2026-10-10.md`). A set-transformer variant registered beside it (LLF-Net) does worse: claim 27 of 60, 22 cells worse than shipped. Trained without one corruption family at a time, LLF holds the no-loss property on 3 of 12 low-light cells and 6 of 12 glare cells it never saw, so what it learned about corruptions does not transfer between them.

LLF-2, learned stream trust. The diagnosis fixed the redesign (`docs/prereg-llf2-2026-10-10.md`): the same pipeline restricted to what a gate does. Its inputs are the box's stream, its confidence (monotone), the strongest support from the other stream (monotone) and a cross-stream duplicate flag (monotone), plus the 16 frame-health features. Within a frame, two boxes of one stream with no cross-stream overlap keep their order. On fresh draws (VIS 945–947, IR 955–957) LLF-2 holds the no-loss property on **52 of 60** cells against 32 for the shipped rule, and with night in training (four capture-order blocks of the night run) on all 30 night cells. It is still not a pass. Clean day sits 0.0073 below VIS alone, the six low-light day cells 0.0068 to 0.0142 below, and against shipped it is worse on 12 cells beyond 0.0060 and better on 26; both registered predictions FAIL (`docs/eval/llf/llf2_2026-10-10.md`).

Deep-feature trust (E7). LLF-2's trust features plus 32 whitened principal components of the frozen backbones' frame embeddings (16 per stream, fitted per system on clean training frames), gradient-boosted trees on the GPU, one model per system and fold (`scripts/deep_gate.py`; Amendment 2 of `docs/prereg-no-retrain-2026-10-10.md`). It meets P1 by the registered rule, 60 of 60 cells, and it is the only system in this paper to do so. The pass is weaker than it reads. Eleven cells sit 0.0060 or more below max(VIS, IR) in point estimate (clean day -0.0088 [-0.0272, +0.0095], clean VIS with IR noise s3 by day -0.0103 [-0.0315, +0.0108]), and they hold because the five per-system models disagree, which widens the intervals to about ± 0.02, not because the fused output reaches the better stream. P2 fails: five cells are worse than shipped beyond 0.0060, the worst clean VIS with IR noise s2 by day (-0.0236 [-0.0430, -0.0041]). Mean fused ship AP over the 60 cells is 0.1485 against 0.0894 for shipped (`docs/eval/no_retrain/deep_gate_fresh.md`). Not adopted.

No gate, and three new gates (E4, E6). The simplest replacement is none: the same streams fused with the veto switched off (UNION). It IMPROVES on shipped by the registered rule, with no cell worse at 0.0014 and 30 better beyond 0.0060 with the interval above zero, and holds the no-loss property on 59 of 60 cells. The exception is clean VIS with IR fog s3 by day, 0.0099 [0.0068, 0.0131] below VIS alone, the support-bonus failure of Appendix D.5. Three new gates registered beside it all lose to it (`docs/eval/no_retrain/gate_zoo_fresh.md`). `selfconf` vetoes a stream when its own top ship score is below 0.10 while the other's is at least 0.25. It is worse than shipped on 18 cells at 0.0014 (glare day -0.0212) and fails the no-loss property on 3. CALUNION maps each stream's scores to the probability of a true positive with isotonic calibrators fitted per system and health state, then concatenates. It is worse on 18 cells (low-light day -0.0342), and isotonic steps tie the boxes they flatten. `proben` adds Bayesian pair fusion [17] on top of it and is worse on 21 cells (-0.0404). Every gate recovers the night cells, as UNION does. None adds anything by day.

A tail cannot hurt (E9). The scorer matches boxes greedily in score order, and its precision envelope is a running maximum from the tail. Boxes appended below every existing box therefore cannot take a match from an earlier box or lower any earlier precision, so AP cannot fall. Two consequences were registered before any number on the fresh draws (Amendment 4). DEMOTE keeps the shipped gate but appends the vetoed stream's boxes below everything instead of deleting them. It is at or above shipped in every system, draw and cell (zero violations above 1e-4), IMPROVES on shipped (24 cells better, none worse), and still fails the no-loss property on 22 cells, because at night IR's boxes lead and VIS's come after them. LEX ranks every VIS ship box above every IR ship box, with no merge, no gate and no parameter. It is at or above VIS by construction, again with zero violations, and it holds the no-loss property on **all 60 cells with no interval below zero** (worst -0.0042 [-0.0104, +0.0021], fogged VIS with noisy IR by day, the one kind of cell where IR is the better stream). It is the only system in this paper to do so other than through wide intervals. It does not IMPROVE on shipped: where IR is healthy, the union's cross-modal support is worth up to 0.0165 by day (clean VIS with IR noise s2), and LEX gives that up. Across the 60 cells LEX averages 0.0037 below UNION, from 0.0170 below to 0.0105 above.

Why frame-level gates fail even when they choose well. AP pools the detections of every frame into one ranking, so a gate that emits two kinds of output also emits two score scales. With clean IR, the union scores the same top boxes 1.13× higher than VIS alone, through the support bonus. On the smoke draw we switched VIS-only output onto a *random* fraction of frames, a choice that carries no information. At three quarters of the frames the result scores 0.3420, below VIS alone (0.3485), below the union (0.3566), and 0.0085 below the interpolation between them. Where the scales agree (IR fog s3, ratio 0.99) the loss is a third as large (`docs/eval/no_retrain/switching_941.md`). An unregistered arm that tiered only the frames where IR was flagged unhealthy lost to both UNION and LEX for the same reason. Every frame-level gate in this project, the shipped veto included, has paid this pooling penalty on top of whatever its decisions were worth. A gate has to keep one score scale across frames, and LEX does so by construction.

F.2 Changes that retrain nothing

Before any of their numbers existed we registered changes that leave every detector frozen (`docs/prereg-no-retrain-2026-10-10.md` and its four amendments, each written before its arms were scored). They are scored on the same fresh

draws as Appendix F.1, against the shipped system of the same seed, with one rule for all of them. An arm IMPROVES if no cell is worse than shipped by 0.0014 with the interval below zero and at least one cell is better by 0.0060 with the interval above zero. Table S12 gives the verdicts.

TABLE S12
NO-RETRAINING CHANGES ON THE FRESH DRAWS (VIS 945–947, IR 955–957), 60 CELLS, FIVE PHASE 3 SYSTEMS, SHIP AP

Arm	What changes	Better ($\geq$0.0060)	Worse ($\geq$0.0014)	Claim fails	Mean fused	Verdict
shipped `crossmodal26m`	—	—	—	28	0.0894	—
UNION	veto off	30	0	1	0.1544	IMPROVES
`selfconf`	gate on each stream's top score	28	18	3	0.1503	not adopted
CALUNION	isotonic P(TP) per health state, concatenated	28	18	9	0.1460	not adopted
`proben`	calunion + Bayesian pair fusion	28	21	8	0.1459	not adopted
LEX	every VIS box above every IR box	29	13	0	0.1507	not adopted
DEMOTE	shipped, vetoed stream kept as the tail	24	0	22	0.1144	IMPROVES
LLF-2, learned stream trust	monotone trust features + frame health	26	21	8	0.1487	FAIL (P1, P2)
deep-feature trust (E7)	LLF-2 + 32 backbone components	28	17	0	0.1485	FAIL (P2)
restoration, gated (E2)	IR NLM where IR health fails, shipped gate	10	2	26	0.0962	not adopted
`ens2`	2 checkpoints per stream, shipped gate	18	0	28	0.0974	IMPROVES
ENS5	5 checkpoints per stream, shipped gate	43	0	28	0.1045	IMPROVES
ENS5_UNION	5 checkpoints, veto off	56	0	1	0.1751	IMPROVES
ENS5_LEX	5 checkpoints, VIS tier above IR	54	0	1	0.1716	IMPROVES
`ens5_restore`	5 checkpoints, gated IR NLM, shipped gate	49	0	27	0.1157	IMPROVES
`full`	5 checkpoints, gated IR NLM, veto off	57	0	1	0.1825	IMPROVES
`tta`	flip TTA, shipped gate	14	0	28	0.0929	IMPROVES
FULL_TTA	`full` + flip TTA	57	0	1	0.1862	IMPROVES
FULL_TTA_LEX	FULL_TTA with tiers instead of union	57	0	2	0.1803	IMPROVES

Better / worse = cells beyond the floor with the interval clear of zero, against shipped. Claim fails = cells where fused $-$ max(VIS, IR) ≤ -0.0060 with the interval below zero, measured within each arm. The ensemble arms share their streams across the five systems, so their claim intervals have zero width: a claim failure there is a point estimate.

Ensembles are the largest lever, and they compose with the gate's removal. Merging the five frozen checkpoints of each stream with plain WBF raises VIS ship AP by +0.0192 on average across the 60 cells (+0.0037 to +0.0316) and IR by +0.0073. Under the shipped gate that gain is real but trapped: ENS5 is better than shipped on 43 cells and worse on none, yet it still fails the no-loss property on the same 28 cells, because the night veto still discards the better stream. With the gate removed, ENS5_UNION is better than shipped on 56 of 60 cells, worse on none, and fails the no-loss property on one (clean VIS with IR fog s3 by day, -0.0102). Its mean fused AP over the 60 cells is 0.1751 against 0.0894 for shipped: clean day 0.3751 against 0.3566, clean night 0.3124 against 0.0687. ENS5_LEX is better than shipped on every one of the 60 cells (the smallest gain +0.0036) and fails the no-loss property on one, fogged VIS with noisy IR by day (-0.0073), the one kind of cell where IR is the better stream. These systems cost five detector passes per stream where shipped costs one, and the gain is a property of the detectors, not of the fusion: ensembling helps every arm it is put under.

Restoration helps only where the restored stream is the one being read. A selection rule declared in the registration picked one classical restoration per modality on the smoke draw (every fourth frame, three systems). It required no clean-cell loss beyond 0.0014 wherever the shipped gate's own degradation flag activates (`docs/prereg-no-retrain-2026-10-10.md`, E2). No VIS method passed: on clean night frames, where the darkness flag activates on every frame, auto-exposure, CLAHE and two dark-channel dehazers cost 0.0021 to 0.0460. For IR, non-local-means denoising passed and was selected (+0.0139 mean gain on the IR stream's corrupted cells). Applied where the IR health flag activates, under the shipped gate, it is better than shipped on 10 cells. The largest gains, +0.0477, are noisy-IR nights, where the shipped output is IR alone. It is worse on 2 cells, clean and low-light VIS with IR noise s2 by day, -0.0064 and -0.0080, although the restored IR stream itself gains +0.0094 there; it does not improve. Horizontal-flip test-time augmentation, merged per checkpoint by the same WBF, raises VIS by +0.0075 on its own (`tta`, 14 cells better) and adds +0.0057 on top of the five-checkpoint ensemble. The composition of everything that helps (five checkpoints per stream, gated IR restoration, flip TTA, no gate; FULL_TTA) is the best system measured in this paper. It is better than shipped on every one of the 60 cells (the smallest gain +0.0049), reaches 0.3801 on clean day and 0.3192 on clean night against shipped 0.3566 and 0.0687, and fails the no-loss property on one cell, clean VIS with IR fog s3 by day (-0.0082). It costs ten detector passes per stream per frame. It was chosen on development frames; its registered test on pohang04, run afterwards, replicated in all nine cells (Appendix G.3).

The shipped constants under v2. The constants of `crossmodal26m` were chosen and re-evaluated on the v1 corruptions. A descriptive audit moved each one at a time on v2, on draws 942–944, against the shipped preset (`docs/eval/corruption_v2/constants_audit.md`); no arm in it replaces a shipped value without a registered test. The fusion constants stand. No setting of the IR capability scale ($\times$1 to $\times$16) or the WBF threshold (0.55 to 0.95) is better than shipped on any cell beyond 0.0060, and the thresholds 0.55, 0.70 and 0.95 are worse on 11, 8 and 10 cells at 0.0014. Switching the support bonus off is better on five fogged-IR day cells (up to +0.0123, clean VIS with IR fog s3, the failure of Appendix D.5) and worse on nine (up to -0.0158), and its softer settings move no cell beyond 0.0060. Turning the IR merge veto on is worse on 11 cells

and better on none, and letting `veil` activate by day is worse on 11 (low-light day −0.1574), so both shipped switches are confirmed. The audit finds one switch that is wrong on v2, and it sits on the closed night axis. With the weak-IR fallback off, a disarmed IR stream can no longer confirm a veto. That is better than shipped on 25 cells and worse on none: five low-light day cells (+0.0094 to +0.1684, more than R1 recovers) and 20 nights with degraded IR, and claim failures fall from 28 to 5. The fallback was built for fogged nights with damaged IR, and on v2 it does not pay there either. Switching the veto off altogether reproduces the fresh-draw UNION result on these draws (30 better, none worse, claim fails 1). The fallback switch meets the audit's registered bar for a test on fresh draws, and we did not register it: it changes the closed night axis, and UNION, already scored on the fresh draws, dominates it here (30 cells better against 25, claim failures 1 against 5).

APPENDIX G: THE HELD-OUT RUN

G.1 Protocol of the look

pohang04 (26,188 VIS images, no IR labels) had never been scored by any fusion system, and no Phase 3 checkpoint or reference cache had seen it (Appendix A.5, which also states what earlier artifacts did see). Its verdict is scored exactly once (its corrupted cells were re-scored once more under the v2 corruptions; Appendix G.2). HOLDOUT-GAP is declared if and only if `AP_ref − AP_p04 ≥ 0.0060` and the 95 percent block-bootstrap interval on that delta lies entirely above zero. `AP_ref = 0.2898 [0.2580, 0.3175]` is the pooled pohang02 + pohang03 development group, deliberately the weaker of the two development references (the pohang00 group is 0.3955 [0.3352, 0.4827]) so that the test asks whether pohang04 is worse than run-to-run variation already observed in development, not worse than the best development number. The spread between the two development groups is 0.1057, larger than the 0.033 expected before Phase 3.

The endpoint is the fused score of the shipped `crossmodal26m` system against the existing VIS labels (26,188 files, 156,652 boxes, 286 empty; hash `c06611a684f4`). No thermal labels exist for pohang04 and none were drawn. All five seed pairs from the Phase 3 retrain are scored, VIS seed k with IR seed k, and the headline is the mean over seeds; no constant is re-tuned after seeing results. All eleven benchmark cells are scored with fresh corruption-draw seeds, but only the clean/clean cell carries the verdict; the other ten are descriptive. Day and night are classified by solar elevation from the GPS timestamp; pohang04 contributes only to day cells.

The look is mechanically single-shot. The scoring script refuses to run unless the repository is at a clean FREEZE commit, all 316 files in the hash manifest (10 checkpoints, 10 reference caches, 190 pohang04 caches, 76 frame-statistic files, and substrate and calibration files) verify, the labels hash correctly, and the development reference reproduces exactly (0.3894193201201913). It writes a `LOOK_TAKEN` marker before scoring begins, so a crash mid-look still counts as the look having been taken. The exposure is logged in the project's ledger.

Under this pre-registration no retrained-versus-deployed comparison is reported (the five-seed design cannot power it), and no Gaussian versus MC-Dropout versus ensemble comparison is reported on pohang04 (the VIS MC and ensemble arms were trained on 9,841 pohang04 frames).

G.2 Result of the look

HOLDOUT-GAP is not declared: the point estimate clears the floor, but the interval does not exclude zero. Clean/clean fused ship AP on pohang04, taken over 12,482 day pairs and averaged over the five seed pairs, is **0.2682 [0.2576, 0.2793]**. The per-seed values are 0.2789, 0.2696, 0.2655, 0.2646 and 0.2625. The delta against `AP_ref` is **D = +0.0216, 95 percent interval [−0.0120, +0.0502]**, from an unpaired moving-block bootstrap (L = 20, n_boot = 1000, bootstrap seeds 0 and 1, differenced replicate by replicate). The point estimate clears the 0.0060 floor and every other reported floor up to 0.0100. The rule is not met solely because the interval does not exclude zero.

This is not a finding of no gap. A gap of 0.0216 was observed; the data exclude neither zero nor a gap as large as 0.05. The interval is wide because the comparison is unpaired across runs and so carries run-level variance, of which this data has a great deal. The two development groups differ from each other by 0.1057, 4.9 times the held-out gap. pohang04 lies just below the lower development group's point estimate and inside its interval [0.2580, 0.3175]: within the run-to-run variation development already shows, not outside it.

What the look does not test.

- The verdict covers clean day only.
- The endpoint is scored against VIS ground truth alone, so a true IR-only detection counts as a false positive.
- Only the fused output was scored; no VIS-only or IR-only arm exists for pohang04. The look therefore does not test whether fusion is at or above the better single stream on held-out data, and the per-stream generalization of either detector may not be claimed.
- pohang04 is all daylight, so the night defect of Appendix D.4 is outside its reach. As recorded behavior of the frozen system, not a tuned quantity, the raw IR night flag activates on 505 of its 12,482 daylight pairs (4.0 percent).
- The Mahalanobis scorer is inert in the shipped preset, so the look does not validate it. Its references were rebuilt from pohang04-free lists before the freeze (Appendix A.5).

TABLE S13
POHANG04, ALL CELLS (DESCRIPTIVE EXCEPT CLEAN/CLEAN). SEED-MEAN FUSED SHIP AP, LOCAL AP, BLOCK-BOOTSTRAP 95% INTERVAL; CORRUPTED CELLS AVERAGE FOUR FRESH DRAWS (VIS 941–944, IR 951–954). THE CORRUPTED CELLS WERE RE-SCORED UNDER THE V2 CORRUPTIONS ON 2026-10-10, A DISCLOSED SECOND EXPOSURE; THE LOOK'S V1 VALUES STAND BESIDE THEM

Cell (VIS / IR)	v2 AP	v2 95% interval	v1 AP (look)
clean / clean (verdict, not re-scored)	0.2682	[0.2576, 0.2793]	0.2682
clean / blur_s2	0.2673	[0.2570, 0.2779]	0.2671
clean / noise_s2	0.2635	[0.2542, 0.2738]	0.2692
clean / fog_s2	0.2666	[0.2565, 0.2771]	0.2691
clean / glare_s2	—	not modeled in v2	0.2668
rain_s2 / clean	0.2209	[0.2107, 0.2324]	0.1850
lowlight / clean	0.2134	[0.2040, 0.2237]	— (new)
fog / clean	0.0445	[0.0393, 0.0505]	0.0580
blur_s3 / clean	0.0419	[0.0389, 0.0448]	0.0418
noise_s2 / clean	0.0123	[0.0097, 0.0142]	0.0114
blur_s3 / glare_s2	—	not modeled in v2	0.0407
lowlight / glare_s2	—	not modeled in v2	0.0121

One clean run does not retroactively create a test set for the development decisions of Appendices D–F.

The descriptive cells carry no pass or fail language. Under v2 they show what the development cells show: the fused output tracks the VIS stream. IR-side corruption leaves it within 0.0047 of clean (v1: $\pm$0.0014), and VIS-side corruption moves it by up to 0.2559. Three cells moved with the corruption model rather than with the system. v2 low light, at the real night exposure, leaves 0.2134 where v1's additive clip had left 0.0121; that v1 cell also carried IR glare, which v2 does not model, so `lowlight / clean` replaces it. Rain, now confined to the image rows, is 0.0359 higher. v2 fog, which loses contrast with distance instead of applying a fixed blur, is 0.0136 lower. The three IR-glare cells have no v2 counterpart and keep their v1 values.

The re-score is a second exposure of pohang04, and we disclose it. After the v1 corruptions were found broken (Appendix B.1), the corrupted cells were re-scored under v2 with the look's systems, preset, calibration, draws, statistic and bootstrap. Only the corruption changed, along with the inference batch of the corrupted caches (4 instead of 1; |Δ ship AP| ~1e-5 measured). The re-score was registered before any v2 pohang04 number existed (`docs/prereg-p04-v2-rescore-2026-10-10.md`). The clean/clean verdict was not re-scored, and the look's markers were verified unchanged before and after. The bootstrap ran on the GPU after being checked bit-for-bit against the CPU path on 96 replicates. pohang04 has still fitted nothing. Every later pohang04 result (Appendix G.3) is a further registered exposure of the same daylight run, logged in the exposure ledger, and is to be weighed as a repeated look, not a first one.

The look ran once, for 18.6 hours, at freeze commit `85a07c1`. It is logged in the exposure ledger, and a tracked marker makes any re-run refuse.

G.3 Later registered scorings

On 2026-10-11 the project decided that pohang04 is scored whenever a result needs held-out validation. Each scoring is registered before its numbers exist and logged in the exposure ledger. pohang04 has still fitted nothing: no constant, threshold or calibration of any system below. It has now been scored thirteen times: the look (Appendix G.2), the v2 re-score of its corrupted cells, and the eleven registrations below (P-U and P-F together, P-T, P-A, P-FT, P-C2, P-G, P-FG, P-C2G, the gate ablation GA with its feature-gap amendment, Appendix G.5, and the gate-free shallow BN-mix GF31, `docs/prereg-gatefree-p04-2026-10-16.md`). Each later result should be weighed as a further registered exposure of one daylight run, not as a first look. All use the look's nine cells (clean/clean and the eight v2 cells; P-G adds a VIS glare cell), the five Phase 3 systems, development calibration, and the look's 1,000 moving-block replicates (L = 20, seed 1). Each Δ is paired.

Gate-free systems (`docs/prereg-p04-heldout-gatefree-2026-10-11.md`).

- **P-U, UNION non-inferior by day: PASS.** No cell is worse. UNION differs from SHIPPED only where the veto activates by day: +0.0045 [+0.0032, +0.0067] on VIS noise and +0.0018 [+0.0006, +0.0033] on low light. Every other cell is identical.
- **P-F, FULL_TTA replicates: REPLICATES.** FULL_TTA is better than SHIPPED in all nine cells, by +0.0083 [+0.0067, +0.0108] (VIS noise) to +0.0325 (rain), with every interval above zero; clean/clean is 0.2925, +0.0243 [+0.0199, +0.0278]. The best development system of Appendix F.2 keeps its direction and most of its size on held-out daylight.
- Descriptive:
  - ENS5_UNION is better in all nine cells, by +0.0083 to +0.0286.
  - LEX is better in eight, by up to +0.0134; its ninth cell, blur s3, is +0.0009 [$-$0.0004, +0.0016].
  - FULL_TTA_LEX scores above FULL_TTA in five cells, with intervals above zero, and below it on blur s3 ($-$0.0016 [$-$0.0027, $-$0.0006]).

TABLE S14
POHANG04, REGISTERED LATER SCORINGS: SHIP AP AGAINST THE VIS GROUND TRUTH, MEAN OVER THE FIVE SYSTEMS (AND THE FOUR DRAWS OF A CORRUPTED CELL)

Cell (VIS / IR)	shipped	UNION Δ	FULL_TTA Δ	ENS5_UNION Δ	LEX Δ	FULL_TTA_LEX Δ	IG-BN − UNION
clean / clean	0.2682	+0.0000	+0.0243	+0.0232	+0.0096	+0.0273	0 (gate never activates)
clean / blur_s2	0.2673	+0.0000	+0.0263	+0.0250	+0.0106	+0.0284	+0.0000
clean / noise_s2	0.2635	+0.0000	+0.0293	+0.0266	+0.0134	+0.0317	0 (gate never activates)
clean / fog_s2	0.2666	+0.0000	+0.0257	+0.0242	+0.0117	+0.0291	0 (gate never activates)
blur_s3 / clean	0.0419	−0.0000	+0.0112	+0.0102	+0.0009	+0.0095	+0.0001
noise_s2 / clean	0.0123	+0.0045	+0.0083	+0.0083	+0.0039	+0.0076	+0.1823
rain_s2 / clean	0.2209	+0.0000	+0.0325	+0.0286	+0.0026	+0.0350	+0.0036
fog / clean	0.0445	+0.0000	+0.0142	+0.0127	+0.0010	+0.0164	+0.0370
lowlight / clean	0.2134	+0.0018	+0.0324	+0.0264	+0.0052	+0.0361	+0.0186

The first five Δ columns are against SHIPPED; the IG-BN column is against UNION. Bracketed intervals for the headline entries are in the text.

- **The claim on held-out data.** SHIPPED and UNION fail it in the four cells with clean VIS, at −0.0085 to −0.0132: on pohang04 the clean VIS stream alone beats the shipped fusion.
  - LEX and FULL_TTA_LEX hold it in all nine cells. Their tier order guarantees at least the VIS stream.
  - FULL_TTA falls short of its own better stream by at most 0.0027 (low light), inside the 0.0060 margin.

Test-time adaptation. The seven adaptation scorings (P-T = IG-BN, P-C2 = IG-BN-causal, P-G = IG-BN+glare, P-C2G = IG-BN-causal+glare, P-FT = IG-BN+ens, P-FG = IG-BN+ens+glare and P-A = IG-BN+AE) are reported in Section V-B (Table IV) and drawn in Fig. S6, and the descriptive TTBN_LEX arm in Section V-D.

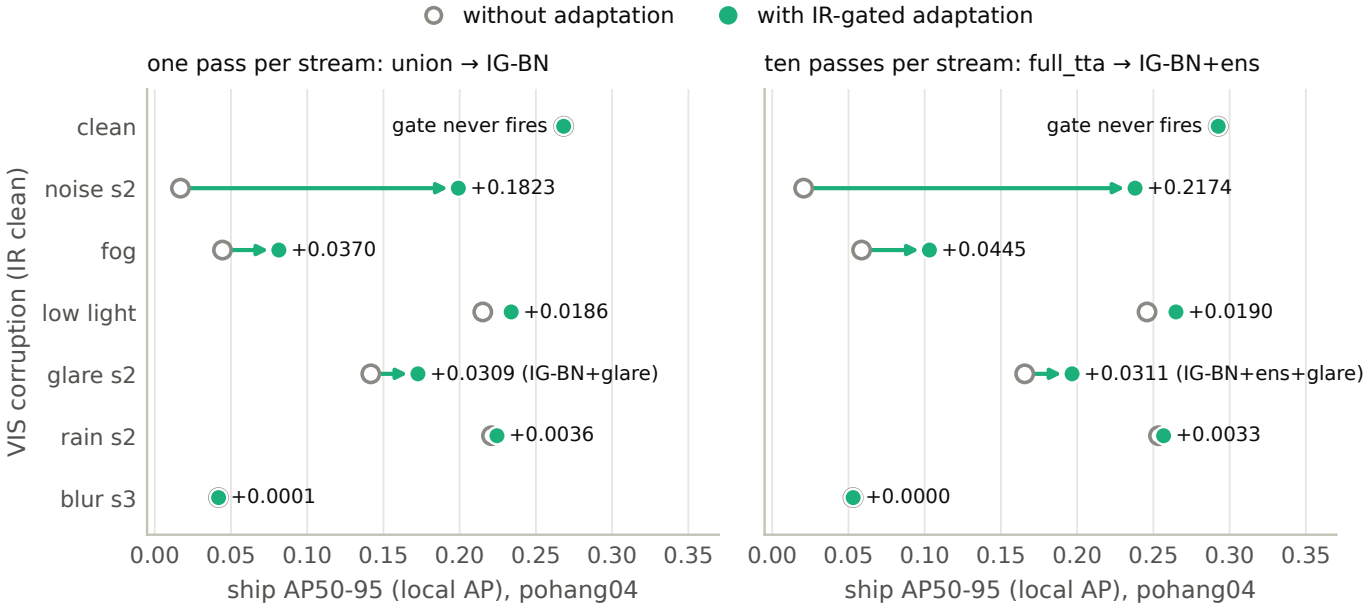


Fig. S6. **IG-BN on the held-out run (pohang04), per cell.** Ship AP against the visible ground truth, mean over the five systems and four draws, thermal stream clean in every row; left, one pass per stream (UNION hollow, IG-BN filled; IG-BN+glare on the glare row); right, ten passes (FULL_TTA and IG-BN+ens; IG-BN+ens+glare on glare). Labels are the paired Δ; intervals are in Table IV.

G.4 A night test on frames no fusion system had seen

Every night number in Appendices D–F comes from development: 1,032 pohang01 frames of the VIS val split. pohang04 is all daylight. pohang01's **test** split, however, had never been fusion-scored: a scan of all 263 detection-cache directories found none of its frames. Its 1,109 pairs whose VIS and IR frames are both test-split frames therefore form a night test for the fusion rules. They are not new for the detectors, because the Phase 3 detectors trained on the same run's train split. We registered the test before building a single cache (`docs/prereg-p01-nighttest-2026-10-11.md`).

Both registered predictions hold.

- UNION beats SHIPPED at night by +0.0273 [+0.0154, +0.0400] (P-N1, REPLICATES).
- UNION keeps the no-loss property, −0.0010 [−0.0049, +0.0031] (P-N2, PASS).
- SHIPPED, which vetoes VIS on 90.3% of these frames, fails the no-loss property by −0.0283.

- LEX is +0.0314 above SHIPPED and holds the no-loss property (+0.0031); ENS5_UNION is +0.0257.
- FULL_TTA was added afterwards as a descriptive amendment, registered before its flipped and restored views existed.
  - It is +0.0287 above SHIPPED, but only +0.0014 [−0.0065, +0.0068] above UNION, against +0.0297 on development night.
  - It is below UNION on one test stretch (0.0355 against 0.0406) and far above it on the other (0.1509 against 0.1095).
  - The ensemble's night gain does not carry over as a pooled number.

The levels do not transfer. VIS ship AP is 0.0516 here, against about 0.25 on development night, and UNION is 0.0506 against 0.2895. pohang01's paired val and test frames come in contiguous stretches of about 52 seconds. Development night is two stretches and the test two others, and VIS ship AP ranges from 0.04 to 0.35 across the four. A night AP level is a property of the episode. The direction of the UNION-over-SHIPPED result and the no-loss property held on both test stretches. The moving-block intervals resample within four episodes and cannot see the variance between them, so they are narrower than an episode-level interval would be. The full table is in `docs/eval/night_p01test_2026-10-11.md`.

G.5 The gate ablation

Gate ablation (GA and its feature-gap amendment). Registered with predictions before any arm was scored (`docs/prereg-gate-ablation-2026-10-14.md`). Every arm reads one complete adapted stream per condition and draw: N4's stored records for the blocks N4 ran and new blocks for the rest, built on N4's code path (development fp32, rebuilt blocks bit-identical to stored ones; held-out fp16, where a rebuild moves a development cell by about 1e-5). The IG-BN arm reproduced N4's registered scores exactly on development and in all 80 held-out jobs where it was recomputed. On the held-out run the sun is up on every frame (solar elevation +56° to +63°), so the solar gate is the ungated arm and the clock gate is the visible term alone. Held-out (the tenth scoring): ungated adaptation is better than UNION in all nine cells and better than IG-BN in eight, from +0.0008 (low light) to +0.0228 (fog), against the registered prediction that it would lose on the clean-visible cells; the thermal day term alone is better in eight and also contradicts its prediction; the visible term alone (= the clock gate) is better than IG-BN under visible noise (+0.0090 [+0.0056, +0.0132]) and equal elsewhere. The feature-gap amendment (the eleventh scoring): the gap gates activate on 98.4–99.9 percent of held-out frames; alone they are better than UNION in all nine cells, like the ungated arm and against their registered prediction, and with the visible term they replicate and tie the clock gate (better or worse on no cell). Of the registered held-out predictions, four failed (ungated, thermal day alone and both gaps alone, all of which gained where a loss was predicted) and the others held. Per-cell values are in the records listed for Table S15 (Table S18).

Which part of the gate does the work. We registered five other gates on the same adaptation, with predictions, before any of them was scored (Table S15): none, the visible term alone, the thermal day term alone (p_{05}^{IR} and the thermal health term), solar elevation alone, and the visible term with solar elevation in place of both thermal terms (the sun from the stereo timestamp and GPS fix, as the held-out look classifies day and night, Appendix G.1). Every registered prediction held on development. Ungated BN-mix is worse than UNION on 18 of 60 cells: every clean-visible cell, −0.0101 [−0.0147, −0.0055] on clean day and −0.0195 [−0.0288, −0.0101] on clean night, and every low-light night cell, down to −0.0233. The visible term alone adapts every night frame, because q_{vis} flags them all (Section IV-A), and is worse on all twelve clean and low-light night cells. The thermal day term alone adapts clean day frames and is worse on clean day (−0.0101) and with light thermal fog (−0.0048). Both halves of the gate are needed. The day half, however, does not need the thermal camera. The visible term with the clock (the clock gate) passes, with 17 cells better than UNION and none worse, and against IG-BN it is worse on no cell and better on 13 of the 15 fogged or low-light day cells with a corrupted thermal stream, where the thermal stream no longer confirms daylight and IG-BN stops adapting. Under thermal noise IG-BN's gate activates on no low-light frame and the clock gate on all of them (+0.0709 [+0.0592, +0.0825] over IG-BN); under thermal fog at severity 3, +0.0660 [+0.0549, +0.0771]. Wherever the thermal stream is clean, the two gates select the same frames. On this development set the solar rule agrees with the run-level day–night split on every frame (day frames at +32.9° to +50.4°, night at −7.2° to −5.2°), so the clock was never asked about dusk, overcast or a covered berth, the cases where it could be wrong.

A model-internal gap does not replace the day term. The visible term is one model-internal signal. A stronger one is the feature-distribution gap of Yoo et al. [18], which adapts a detector only when the running mean of its test features drifts from the training features by more than an in-domain gap; night frames are in the training distribution, so it might stay quiet at night without a second sensor or a clock. We registered it as an amendment before scoring it: the pooled head-input features each cache already stores, training statistics from 3,181 training-split frames (day and night), their EMA momentum and thresholds, and a second arm with a bound fixed at the maximum over the reference frames, each alone and with the visible term. It does not stay quiet. The gap gate activates on 90–100 percent of development night frames and on 67–99 percent of the night-test pairs (Section V-C), so all four arms fail on every clean and low-light night cell (clean night −0.0195 [−0.0288, −0.0101] with Yoo et al.'s thresholds), and with the visible term they are worse than the clock gate on 12 and 18 cells and better on none (Table S15). It also activates on 61–99 percent of clean day frames. A stream of video frames from one run, day or night, sits farther from the pooled day-and-night training mean than two random halves of the training frames sit from each other, so on this detector the gap measures distance from the training mixture, not degradation.

TABLE S15
GATE ABLATION: THE SAME ADAPTATION UNDER TEN GATES, SHIP AP, PAIRED AGAINST UNION AND AGAINST IG-BN

Gate (adapt where)	Dev better / worse vs UNION	Dev verdict	Dev better / worse vs IG-BN	Held-out better / worse vs UNION	Held-out better / worse vs IG-BN
none (every frame)	23 / 18	FAIL	20 / 18	9 / 0	8 / 0
$q_{\mathrm{vis}} < 1$	17 / 12	FAIL	13 / 12	3 / 0	1 / 0
thermal day $\wedge$ thermal healthy	9 / 2	FAIL	5 / 2	8 / 0	6 / 0
sun above horizon	23 / 6	FAIL	20 / 6	= none	= none
$q_{\mathrm{vis}} < 1 \wedge$ sun above horizon	17 / 0	PASS	13 / 0	= $q_{\mathrm{vis}} < 1$	= $q_{\mathrm{vis}} < 1$
feature gap, Yoo et al.'s rule	23 / 18	FAIL	20 / 18	9 / 0	8 / 0
feature gap, calibrated bound	23 / 18	FAIL	20 / 19	9 / 0	8 / 0
$q_{\mathrm{vis}} < 1 \wedge$ feature gap (Yoo)	17 / 12	FAIL	13 / 12	3 / 0	1 / 0
$q_{\mathrm{vis}} < 1 \wedge$ feature gap (calibrated)	17 / 12	FAIL	13 / 13	3 / 0	1 / 0
IG-BN	5 / 0	PASS	—	3 / 0	—

Development: 60 cells, five systems, draws 942–944, t-intervals; held-out: the nine cells of Table IV, moving-block intervals. Better and worse count cells beyond the rule's floors (Section III-D). On the held-out run the sun is up on every frame, so the solar gate is the ungated arm and the visible term with the clock is the visible term alone.

On the held-out run, adapting every frame wins. The registration predicted that ungated adaptation, and the thermal day term alone, would lose on the clean-visible held-out cells as they did on development. Both predictions fail. On pohang04, adapting every frame is better than UNION on all nine cells and better than IG-BN on eight. It gains +0.0210 [+0.0185, +0.0242] on clean/clean and +0.0202 to +0.0216 on the other clean-visible cells, and over IG-BN it adds +0.0228 under fog and +0.0191 under rain, where IG-BN's gate passes 68 and 19 percent of frames. The development frames come from the runs the detectors were trained on; pohang04 is a run they never saw. On it, a clean visible frame still differs from the training statistics by enough for re-estimation to help, yet q_{vis}, fitted on development, flags only 0.3 percent of those frames. So the visible term, which development shows is needed, withholds about 0.02 ship AP per held-out cell. The clock again matches or beats the thermal terms. On this run the clock gate is the visible term alone, and it beats IG-BN under visible noise by +0.0090 [+0.0056, +0.0132], where the thermal health term withholds 15 percent of frames; elsewhere it equals IG-BN. The held-out run cannot test the day half of the gate, because the sun is up on every frame. The feature-gap gates activate on 98.4–99.9 percent of held-out frames, so alone they track the ungated arm (better than UNION in all nine cells, against their registered prediction), and with the visible term they tie the clock gate (better or worse on no cell). On daylight data the gap cannot fail the way it fails at night.

APPENDIX H: BACKBONE BENCHMARK

Ninety-three training runs, 31 YOLO variants with three seeds each from six families (YOLOv8, [19]; YOLOv9, [20]; YOLOv10, [21]; YOLO11, [22]; YOLO12, [23]; YOLO26, [12]), were trained on the restored labels of Appendix A.4 and stopped by one rule: 20 epochs without a new best mAP50-95. The grid first trained 25 epochs on the training server. Each run was then continued from its 25-epoch weights with the early stopper seeded to keep counting from the base run's best epoch, so a run that had already spent 15 of its 20 epochs stopped after five more without a new best. Each run is scored at its best epoch, the earliest maximum, which is the checkpoint the stopper keeps.

The top three are not separable. yolo26x (0.2666 $\pm$ 0.0040), yolo26l (0.2664 $\pm$ 0.0070) and yolo26m (0.2626 $\pm$ 0.0109) differ by at most 0.0040, below the larger seed sd of every pair, the screening rule used for Phase 1; with $n = 3$ it is a screening rule, not a test. The nominal leader's margin over yolo26l is 0.0002. The next variants, yolov10x (0.2524) and yolo12x (0.2523), sit 0.0142 below the leader, outside both seed sds, but only 0.0102 below yolo26m, inside its sd of 0.0109; two of the three seeds of each are † rows. What the table supports is a YOLO26 m/l/x tier at the top with no resolved order inside it. Within YOLO26, mAP50-95 rises from n (0.2384) through s (0.2502) to m (0.2626) and is flat from m to x (+0.0040).

The capacity floor holds for every family but YOLO26. The five other nano- and tiny-scale models land between 0.1777 (yolov9t) and 0.1943 (yolo12n). yolo26n scores 0.2384, 0.0441 above the best of them and level with the medium models of four other families (yolov9c 0.2377, yolov9m 0.2371, yolov10m 0.2366, yolo12m 0.2366).

Training to patience barely reorders the field. Against the same runs' best within 25 epochs, the variant ranking correlates at Spearman ρ = 0.948 over the 24 variants whose continuations are all logged (0.967 over all 31, where † rows equal their 25-epoch values by construction). The top five are unchanged, and 19 of the 79 logged runs beat their 25-epoch best. The exception is yolo26n: all three seeds improved, +0.0276 on the mean, with best epochs near 63, moving it from 25th to 15th. A 25-epoch budget penalized it specifically. The next-largest gains are yolov9c (+0.0079), yolov8x (+0.0073) and yolo26m (+0.0051).

TABLE S16
VIS BACKBONE BENCHMARK AT PATIENCE 20, SEED MEAN ± SD

#	Variant	n	mAP50-95	mAP50	25 epochs	Δ	Best epoch
1	yolo26x	3 (1†)	0.2666 ± 0.0040	0.668 ± 0.033	0.2666 ± 0.0040	+0.0000	14
2	yolo26l	3	0.2664 ± 0.0070	0.677 ± 0.003	0.2655 ± 0.0056	+0.0010	20
3	yolo26m	3	0.2626 ± 0.0109	0.672 ± 0.026	0.2576 ± 0.0056	+0.0051	19
4	yolov10x	3 (2†)	0.2524 ± 0.0051	0.642 ± 0.010	0.2524 ± 0.0051	+0.0000	16
5	yolo12x	3 (2†)	0.2523 ± 0.0091	0.641 ± 0.017	0.2519 ± 0.0088	+0.0004	21
6	yolo26s	3	0.2502 ± 0.0090	0.665 ± 0.007	0.2472 ± 0.0050	+0.0030	27
7	yolov10b	3	0.2476 ± 0.0051	0.623 ± 0.015	0.2476 ± 0.0051	+0.0000	20
8	yolov8l	3	0.2470 ± 0.0033	0.634 ± 0.006	0.2470 ± 0.0033	+0.0000	11
9	yolo11x	3 (2†)	0.2466 ± 0.0078	0.650 ± 0.022	0.2466 ± 0.0078	+0.0000	16
10	yolov8x	3	0.2457 ± 0.0039	0.619 ± 0.014	0.2384 ± 0.0027	+0.0073	40
11	yolo12l	3 (2†)	0.2444 ± 0.0061	0.633 ± 0.016	0.2444 ± 0.0061	+0.0000	15
12	yolov10l	3 (2†)	0.2444 ± 0.0078	0.625 ± 0.007	0.2434 ± 0.0060	+0.0010	31
13	yolo11m	3	0.2427 ± 0.0038	0.645 ± 0.025	0.2427 ± 0.0038	+0.0000	16
14	yolo11l	3	0.2414 ± 0.0092	0.631 ± 0.010	0.2406 ± 0.0089	+0.0008	19
15	yolo26n	3	0.2384 ± 0.0064	0.653 ± 0.016	0.2107 ± 0.0049	+0.0276	63
16	yolov9c	3	0.2377 ± 0.0067	0.608 ± 0.018	0.2298 ± 0.0014	+0.0079	43
17	yolov9e	3 (3†)	0.2376 ± 0.0081	0.600 ± 0.031	0.2376 ± 0.0081	+0.0000	17
18	yolov9m	3	0.2371 ± 0.0117	0.630 ± 0.025	0.2371 ± 0.0117	+0.0000	9
19	yolov10m	3	0.2366 ± 0.0128	0.611 ± 0.018	0.2340 ± 0.0090	+0.0026	39
20	yolo12m	3	0.2366 ± 0.0075	0.616 ± 0.024	0.2346 ± 0.0079	+0.0019	28
21	yolov8m	3	0.2331 ± 0.0039	0.612 ± 0.027	0.2331 ± 0.0039	+0.0000	16
22	yolo12s	3	0.2228 ± 0.0079	0.613 ± 0.026	0.2228 ± 0.0079	+0.0000	13
23	yolov8s	3	0.2202 ± 0.0022	0.605 ± 0.005	0.2202 ± 0.0022	+0.0000	13
24	yolo11s	3	0.2200 ± 0.0022	0.601 ± 0.017	0.2200 ± 0.0022	+0.0000	15
25	yolov10s	3	0.2130 ± 0.0101	0.585 ± 0.036	0.2130 ± 0.0101	+0.0000	12
26	yolov9s	3	0.2052 ± 0.0069	0.573 ± 0.030	0.2007 ± 0.0070	+0.0045	28
27	yolo12n	3	0.1943 ± 0.0054	0.562 ± 0.016	0.1943 ± 0.0054	+0.0000	13
28	yolo11n	3	0.1935 ± 0.0057	0.561 ± 0.018	0.1935 ± 0.0057	+0.0000	9
29	yolov10n	3	0.1913 ± 0.0055	0.554 ± 0.013	0.1913 ± 0.0055	+0.0000	14
30	yolov8n	3	0.1833 ± 0.0028	0.547 ± 0.009	0.1833 ± 0.0028	+0.0000	12
31	yolov9t	3	0.1777 ± 0.0113	0.496 ± 0.021	0.1732 ± 0.0054	+0.0045	33

Ultralytics 8.4.90 validation mAP over ship and buoy, read from the training logs. This is not local AP, so no value here is comparable with any other table (Appendix C.4). Restored labels (train-label hash `8ed69b5974ed`), stride-4 train split (24,070 frames), full VIS validation split (11,352 frames, pohang00–04 including the night run). The best epoch is selected on the split it is reported on, a small optimism shared by every row; this is not a held-out claim. *25 epochs* is the same runs' best within their first 25 epochs; *Best epoch* counts from epoch 1.

† Seeds whose continuation logs are not in the repository (14 runs). They are recorded as reaching the stopping rule with no epoch above their 25-epoch best, so each scores its 25-epoch best row and contributes $\Delta = 0$ by construction. yolov9e seed 0 has 3 of its 7 continuation epochs logged, none above its 25-epoch best.

The selection of yolo26m stands. yolo26m was chosen before this benchmark, on the Phase 1 grid (Table S17), under a rule fixed in advance: the top mAP50-95 tier, then a DFL head, then the simplest fork, then throughput. Table S16 puts yolo26m inside the leading tier again, and no YOLO26 variant has a DFL head, so the rule again falls to throughput. There yolo26m runs at 57.0 FPS per detector against 30.7 for yolo26x, 28.5 versus 15.3 two-stream, a factor of 1.86 (Fig. S7). Its cost against the nominal leader is 0.0040, equal at four decimals to yolo26x's own seed sd and below yolo26m's. yolo26l, now second, runs at 44.9 FPS in the same sweep, slower than yolo26m, so the tie-break is unchanged. Throughput is fp32 at batch 1 with the GPU clock pinned at 1500 MHz, detector `predict()` only. The clock is pinned because without it the ranking is not stable: an earlier sweep with the clock unpinned measured yolo12x (200.3 GFLOPs) faster than yolo12l (89.7), 36.1 against 29.7 FPS, and yolo11x faster than yolo11l, orderings no architecture explains. Locked with `nvidia-smi --lock-gpu-clocks`, the clock read 1500 MHz on all 184 measurements of the sweep (GPU at 50–57 °C), recorded per measurement in the `gpu_clock_mhz` column of `phase1_benchmark/fps.csv`, and both orderings come out as size predicts (yolo12l 34.0 against 23.8, yolo11l 42.0 against 29.0). The unpinned sweep (`phase1_benchmark/fps_seed0_only.csv`) is quoted nowhere else. The two-stream figures are derived, not measured: they halve the single-detector time and exclude the σ head, the image statistics and the decision layer. Measured end to end, the shipped two-stream pipeline runs at 9.1 FPS, bound by its image statistics, not its detectors. They rank backbones; they are not the system's end-to-end throughput.

Disclosures for Table S16. The 25-epoch grid ran on one A100 MIG 3g.40gb slice. Ten continuations ran there before access was lost and 68 on the laptop of Appendix I (RTX 4080, 12 GB). One run (yolo26x seed 1) needed none, because its 25-epoch run had already spent its patience; the remaining 14 are the † rows. All logged runs used Ultralytics 8.4.90. On the laptop the batch was planned per family to fit memory (16, 8 or 4) with a nominal batch of 64, so the effective batch stays 64; one run (yolov9c seed 1) dropped from 16 to 8 mid-continuation after the driver spilled memory to system RAM. Loader workers were 6 on the laptop against 2 on the server. yolov8s seed 2 diverged in the 25-epoch grid and is replaced by seed 3. One server log (yolo12x seed 0) lacks its final continuation row; that run stopped exactly 20 epochs after its best, so the missing

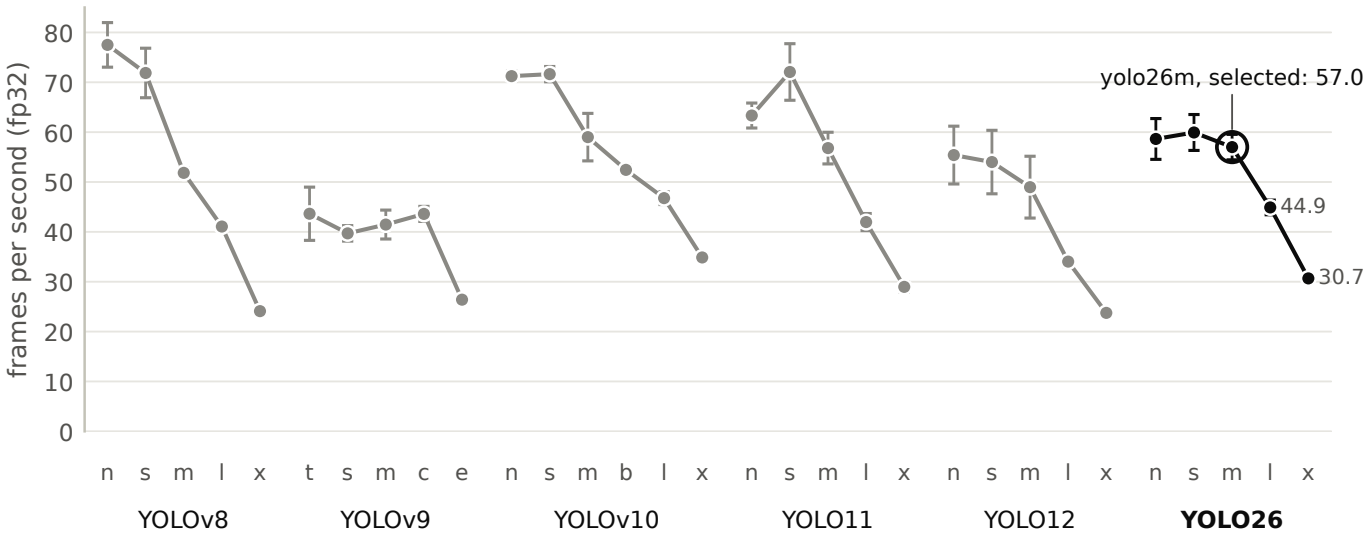


Fig. S7. **Batch-1 throughput of the 31 Phase 1 variants, one size ladder per family, fp32, seed mean ± sd.** One detector's `predict()` per frame at 640 × 640, 500 timed frames after 50 warm-up, GPU clock pinned at 1500 MHz for consistent timing on the laptop of Appendix I (RTX 4080 Laptop); three seeds per variant (two for yolo12x). The sd is measurement precision, not architectural variation: the seeds of a variant share its architecture. The fusion runs one detector per modality, so two-stream throughput is half these values (Table S17), before the σ head, image statistics and decision layer, which are not timed. yolo26m, ringed, is the selected backbone.

row holds no new best. For every logged run, a replay of the stopper over its logged history stops on the run's recorded length.

TABLE S17
PHASE 1 BACKBONE BENCHMARK, THE SELECTION RECORD, MAP50-95 SEED MEAN ± SD

Variant	n	mAP50-95	FPS single / two-stream (fp32, batch 1, clock pinned 1500 MHz; detector `predict()` only, two-stream = single ÷ 2)
yolo26x	3	0.3049 ± 0.0020	30.7 / 15.3
yolo26m	3	0.3016 ± 0.0050	57.0 / 28.5
yolo12x	2	0.3007 ± 0.0049	23.8 / 11.9
yolo26l	3	0.2998 ± 0.0026	44.9 / 22.4
yolo12m	3	0.2906 ± 0.0046	49.0 / 24.5
yolo12l	3	0.2870 ± 0.0096	34.0 / 17.0
yolo26s	3	0.2813 ± 0.0045	59.9 / 30.0
yolo12s	3	0.2783 ± 0.0109	54.0 / 27.0
yolo26n	3	0.2540 ± 0.0058	58.6 / 29.3

Main campaign, local AP, ship class; stride-2 VIS split, pre-restore filtered labels (Appendix A.4), 100 epochs with patience 20 (no run reached 100). Labels, train split, classes, AP convention and epoch budget all differ from Table S16, so no value is comparable across the two tables.

On the Phase 1 grid, pooled over the main and pilot campaigns (Table S17 lists the main campaign only), the top eight variants spanned 0.0055 mAP50-95 against seed standard deviations of 0.0010–0.0066 and were not separable. Pooled, the nominal leader's margin over the next variant was 0.0016, below its own seed sd of 0.0020 (within the main campaign alone it was 0.0033), and the ordering was not stable across metric-reading conventions. Every nano- and tiny-scale model across five families landed between 0.2486 and 0.2567. yolo26m was selected at −0.0033 against yolo26x, a cost that exceeded yolo26x's own sd and was disclosed as the soft spot of the decision.

Disclosures for Table S17. One row (yolo26m seed 0) was trained under library 8.4.7 and re-scored under 8.4.90 (0.3452 to 0.3061); it is tagged and never compared bare. A second, pilot campaign of 66 rows used a split whose machine was later wiped, so cross-campaign contamination cannot be quantified; only yolo12s ran in both (0.2783 versus 0.2810, inside seed sd). Batch size varied 8–32 across machines; the bounded effect is +0.0017, below the smallest seed sd. One row is inadmissible because a resume bug let it train only 14 epochs past its own peak against a required 20. Reported training time (266.8 h) is a lower bound because resumed segments were not summed. Slicing the 27 archived checkpoints by day and night without retraining shows night AP of exactly 0.0000 on all 27 (these predate the restore in Appendix A.4) and day-only and pooled rankings agreeing on all top-three positions, so the selection stood.

A 44-of-93-run IR architecture ladder was stopped early on the basis of an ANOVA ($F(12,26) = 1.037$, $p = 0.447$) that was misread as evidence of equivalence. The minimum detectable spread at that design was 0.02067 against an observed spread of 0.01193, and the Tukey HSD interval on the largest gap is [−0.00314, +0.02700]. The correct statement is that the ladder could not resolve architecture differences, not that the architectures are equivalent. The stop stands on other grounds: the architecture

was frozen before the queue was created, and the ladder trained two classes while the deployed IR configuration is single-class with a P2 neck that was never in the ladder.

APPENDIX I: REPRODUCIBILITY AND IMPLEMENTATION NOTES

Machines. All Phase 3 training ran on one laptop (RTX 4080, 12 GB, Windows) with a CUDA-probed system interpreter; the repository's own virtual environment carries CPU-only torch, and frame statistics must run under it because its numeric libraries reproduce the development statistic files bit-exactly (the GPU interpreter's float32 sums differ by up to 4.4e-7). The v2 corrupted-frame statistics are the exception: they are computed in the cache-build pass, under the GPU interpreter, on exactly the frames the detector saw, because a replay under the other interpreter's image library would not reproduce those frames. The two interpreters differ by at most 6.3e-7 relative on float statistics and not at all on percentiles, and the closest of 232,128 v2 frame statistics to a gate threshold is 1.3e-5 relative away, 20 times that difference, so no gate decision depends on the interpreter (`docs/eval/corruption_v2/gate_threshold_proximity.txt`). Phase 1 ran on an A100 MIG 3g.40gb slice (40,320 MiB, 60 SMs) with Jupyter-only access; it is about 4.5 percent faster than the laptop at steady state, and two earlier contrary measurements were retracted as artifacts. Windows pages rather than raising out-of-memory on an over-large batch (a silent 17× slowdown), so batch ceilings were probed by measured throughput: IR 14, VIS 12 by the probe, and 12 was kept for both in Stage 2 for pre-registration consistency even after a production run measured batch 16 as 16 percent faster.

Gates. A CPU smoke suite runs before any GPU time: benchmark, Gaussian head on the DFL family, Gaussian head on the end-to-end family (both must pass, since σ rides different branches), the UQ pipeline, and Phase 3. A dataset gate re-implements split-fingerprint, leakage, night-filter-hash and balance checks independently of the package (13 checks) and compares train-label content hashes across machines. A contamination gate asserts that pohang04 is absent from every training-consumed list; its first version silently loaded zero images while reporting a pass, and the lesson that a gate verifying the wrong property is worse than no gate is recorded.

Identity. Cache loads validate payload shape, frame count, image paths and detection arrays; pairing is checked by content, not length; a label fingerprint supplements the filename-only split fingerprint (VIS `ae7fa57efb2b`, IR `5fd58f37c799`); a recipe fingerprint over ten training knobs including a content hash of the weights file prevents a completed-run lookup from matching a different recipe. Hashing 107,627 VIS label files costs 15–24 s warm per grid launch.

Queue. Long unattended GPU queues failed in ways worth recording: a paused queue that did not exit its process, so a second runner woke both; zombie CUDA contexts holding memory while the device reported no processes (31 runs lost); a divergence watcher whose window filled exactly at the learning-rate warm-up peak and killed a healthy run (fixed, and verified to change exactly one verdict over 20 historical runs); and a genuine Stage 2 divergence at epoch 6 correctly caught by requiring two consecutive epochs of rising validation loss while training loss still fell. Results CSVs must never be opened in a spreadsheet application, which truncated floats and mangled a version string into a date.

Code. The σ head, fusion and veto, temporal filters, preset assembly, AP and bootstrap, COCO-parity harness, identity checks, label-path resolution, grid driver, and the freeze and look scripts for Appendix G are in the public repository github.com/Laksh-saroha/maritime-tta. Every benchmark row carries a split fingerprint and class tag so that a re-split or filter change cannot be silently averaged against older rows. Phase 1 archives were verified file-by-file (696/696 and 2,281/2,281 with CRC32 and SHA-256 spot checks) before the original tarballs were deleted.

Sources. Every number in a table or figure is read from a result record in the repository; Table S18 lists the record behind each. The records hold the per-cell, per-seed and per-draw values, the bootstrap intervals, and the commands that produced them.

TABLE S18
SOURCE RECORDS OF THE TABLES AND FIGURES READ FROM RESULT FILES

Item	Source records
Fig. 2	scripts/fig_tta_detections.py; frame, crop and per-panel counts in docs/figures/fig_tta_detections.json
Table I	bench_patience20_2026-10-08/table1_ext.md; throughput phase1_benchmark/fps.csv (fp32 rows)
Table III	ttbn/ttbn_dev_2026-10-11.md, ttbn/ttbn_variants_dev_2026-10-11.md, ttbn/ttbn_causal2_dev_2026-10-11.md, ttbn/ttbn_restore_dev_2026-10-11.md, ttbn/ttbn_fulltta_dev_2026-10-11.md, ttbn/ttbn_fulltta_glare_dev_2026-10-12.md
Table S15	ttbn/gate_ablation_dev_2026-10-14.md, ttbn/gate_ablation_p04_2026-10-14.md, ttbn/gate_ablation_gap_dev_2026-10-14.md, ttbn/gate_ablation_gap_p04_2026-10-14.md; night-test rates ttbn/gate_night_p01_2026-10-14.md; registered in docs/prereg-gate-ablation-2026-10-14.md
Table IV	ttbn/ttbn_p04_2026-10-11.md, ttbn/ttbn_causal2_p04_2026-10-11.md, ttbn/ttbn_fulltta_p04_2026-10-11.md, ttbn/ttbn_glare_p04_2026-10-12.md, ttbn/ttbn_causal2_glare_p04_2026-10-14.md, ttbn/ttbn_fulltta_glare_p04_2026-10-12.md
Fig. S6	as Table IV, with holdout_p04_gatefree_2026-10-11.json for the clean baselines; drawn by scripts/paper_figures.py
Table S4	clean p3_night_check_2026-09-27.md; corrupted, severity 2 p3_corrupt_cells_v2_2026-10-10.md; fog severity 1 p3_corrupt_cells_fog_s1_v2_2026-10-10.md; v1 p3_corrupt_cells_2026-09-27.md, p3_fog_s1_2026-09-27.md
Fig. S2	p3_night_check_2026-09-27.json, p3_corrupt_cells_v2_2026-10-10.json
Fig. S3	docs/experiment-log-2026-09-02.md section 7.2 (scored by scripts/eval_night_restore.py); bands fixed in docs/prereg-night-label-restore.md
Fig. S4	scripts/fig_detections.py; frames, crops, flags and IoUs in docs/figures/fig_detections.json
Table S5	corruption_v2/ir_hazards_v2.md
Table S6	uq_day_night_slice_u2_nanpolicy_2026-09-09.md; per class table2_per_class_2026-10-08.md; registered in docs/prereg-uq-day-night-slice.md, Stage B in docs/prereg-uq-day-night-slice-u2-stageb.md
Fig. S5	uq_day_night_slice_u2_nanpolicy_2026-09-09.md
Table S7	uq_mechanism_ablation_ship_2026-10-08.md sections 1–2
Table S8	uq_mechanism_ablation_ship_2026-10-08.md section 3 (scripts/diag_rd1_ship_vs_s0.py)
Table S9	uq_mechanism_ablation_ship_2026-10-08.md section 1, which reproduces both recorded tables; the crossmodal26m record is uq_mechanism_ablation_26m_2026-09-27.md
Table S12	no_retrain/ensembles_fresh.md, no_retrain/gate_zoo_fresh.md, no_retrain/deep_gate_fresh.md, restore/restore_2026-10-10.md, llf/llf2_2026-10-10.md
Table S13	v2 holdout_p04_v2_rescore_2026-10-10.md; v1 holdout_p04_look.md
Table S14	holdout_p04_gatefree_2026-10-11.md, ttbn/ttbn_p04_2026-10-11.md
Table S16	bench_patience20_2026-10-08/table1_ext.md; per run bench_patience20_2026-10-08/table1_ext_runs.csv
Fig. S7	phase1_benchmark/fps.csv

Paths are under docs/eval/ unless they begin with docs/, scripts/ or phase1_benchmark/, in which case they start at the repository root.

APPENDIX J: FURTHER LESSONS FROM THE RECORD

Fusion at this registration quality is aggregation. With a 3–6 px median residual and within-run drift, 0.05 percent of VIS boxes have an IR partner at the adopted threshold. The system does not measure sensor agreement; it concatenates. Stage 1 closed the obvious repair: relaxing correspondence and letting σ arbitrate the merges is non-inferior on one of four conditions. Any claim about cross-modal agreement is out of reach of this system, and the design assumption that decision-level fusion tolerates misalignment was wrong as stated; decision-level fusion avoids the residual by almost never merging.

A corruption is a claim about the camera. The off-the-shelf fog we first used was a fixed 21-px blur, and the gate's fog detector was validated on it: veil was 100 percent correct on that fog and activates on 9 percent of fogged day frames once fog is attenuation with distance. Replacing the corruptions with a camera-chain model moved the night veto from right to wrong under low light, opened a hole in the IR switch's weak-IR fallback that costs up to 0.16 AP, showed the cross-modal support bonus turning against the system once IR is fogged, and took the score-path pass of R-D1 below its floor on corrupted frames. A gate tuned or validated on synthetic corruptions inherits their physics. Before a corruption grades a gate, it should be checked against real frames of the same kind, through the statistics the gate itself reads; that check is what found a defect in our own second model, whose night glare lifted the dark floor of every frame (Appendix B.1, docs/eval/corruption_v2/README.md).

Redundancy is worth what it is independent of. Temporal persistence and view agreement carry no signal beyond confidence; cross-modal agreement carries some; σ carries more. Any future fusion term should be screened for lift against confidence before it is scored in a benchmark.

Synthetic ladders cannot find reference contamination. The Mahalanobis scorer trusted real night because the night run was in its fit set, while distrusting synthetic low-light that looked the same. No severity sweep exposes a statistic that is wrong about the reference population; only a falsification test with the population changed does.

Checkpoint selection is noisier than seeds. A best-epoch maximum over a noisy curve rewards noise, and it inverted the ranking of uncertainty methods and the ranking of machines. Comparisons of uncertainty methods should be carried by calibration metrics, where separations are large relative to this noise, and every headline number should report the epoch-mean alongside.

Small deltas on video need a floor and a block. Without a magnitude floor a gate degenerates into a sign test on 1e-5; without block bootstrapping intervals are half their true width. Applying both corrections retroactively moved 20 of 74 findings to indeterminate. The findings that survived are the large ones, which is the pattern one should expect and want.

The metric is part of the registration. R-D1 registered ship AP and was scored on a macro that included a class only one stream can detect. The macro halved the ship deltas where that class vanished and mixed in a loss on it where it did not, and it returned the opposite verdict for one of two paths. Checking that the scored quantity is the registered one belongs in the same audit as the floor and the interval.

One held-out run buys one number. The pre-registered look produced a point gap of 0.0216 and an interval too wide to say whether it is real. That is the honest yield of a single untouched run on data whose development runs already differ by 0.1057. A generalization claim on this dataset would need more held-out runs, not a better statistic. The registered scorings that followed bought directions, not a test set. Each later system was scored on the same daylight run, so its agreement with development shows replication on one run, not generalization across runs.

Every element was registered before it was scored and replicated on the held-out run (Appendix G.3). The night half of the first rests on the night test of Appendix G.4. The recommendation does not rest on two things. Its gains are on simulated corruptions: the gate does not activate on clean held-out frames, so no real degradation has tested it. And ranking visible boxes above infrared ones scored higher on the held-out run but lower on development; it is a candidate for the next registration, not part of the recommendation. The cost of IG-BN and IG-BN+ens was not timed.

REFERENCES

[1] A. Buslaev, V. I. Iglovikov, E. Khvedchenya, A. Parinov, M. Druzhinin, and A. A. Kalinin, "Albumentations: fast and flexible image augmentations," *Information*, vol. 11, no. 2, p. 125, 2020.

[2] C. Sakaridis, D. Dai, and L. Van Gool, "Semantic foggy scene understanding with synthetic data," *International Journal of Computer Vision*, vol. 126, no. 9, pp. 973–992, 2018.

[3] L. Yang, B. Kang, Z. Huang, Z. Zhao, X. Xu, J. Feng, and H. Zhao, "Depth Anything V2," in *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*, 2024, arXiv:2406.09414.

[4] D. Hendrycks and T. Dietterich, "Benchmarking neural network robustness to common corruptions and perturbations," in *International Conference on Learning Representations (ICLR)*, 2019, arXiv:1903.12261.

[5] H. R. Künsch, "The jackknife and the bootstrap for general stationary observations," *The Annals of Statistics*, vol. 17, no. 3, pp. 1217–1241, 1989.

[6] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, and C. L. Zitnick, "Microsoft COCO: common objects in context," in *Computer Vision – ECCV 2014*, ser. Lecture Notes in Computer Science, 2014, pp. 740–755.

[7] F. Küppers, J. Kronenberger, A. Shantia, and A. Haselhoff, "Multivariate confidence calibration for object detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2020, pp. 1322–1330.

[8] E. Ilg, Ö. Çiçek, S. Galesso, A. Klein, O. Makansi, F. Hutter, and T. Brox, "Uncertainty estimates and multi-hypotheses networks for optical flow," in *Computer Vision – ECCV 2018*, ser. Lecture Notes in Computer Science, 2018, pp. 677–693.

[9] Y. Geifman, G. Uziel, and R. El-Yaniv, "Bias-reduced uncertainty estimation for deep neural classifiers," in *International Conference on Learning Representations (ICLR)*, 2019, arXiv:1805.08206.

[10] N. Bodla, B. Singh, R. Chellappa, and L. S. Davis, "Soft-NMS: improving object detection with one line of code," in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2017, pp. 5562–5570.

[11] R. Solovyev, W. Wang, and T. Gabruseva, "Weighted boxes fusion: ensembling boxes from different object detection models," *Image and Vision Computing*, vol. 107, p. 104117, 2021.

[12] G. Jocher, J. Qiu, M. Liu, S. Lyu, F. C. Akyon, and M. E. Kalfaoglu, "Ultralytics YOLO26: unified real-time end-to-end vision models," 2026, arXiv:2606.03748.

[13] M. Seitzer, A. Tavakoli, D. Antic, and G. Martius, "On the pitfalls of heteroscedastic uncertainty estimation with probabilistic neural networks," in *International Conference on Learning Representations (ICLR)*, 2022, arXiv:2203.09168.

[14] K. Lee, K. Lee, H. Lee, and J. Shin, "A simple unified framework for detecting out-of-distribution samples and adversarial attacks," in *Advances in Neural Information Processing Systems 31 (NeurIPS 2018)*, 2018.

[15] O. Ledoit and M. Wolf, "A well-conditioned estimator for large-dimensional covariance matrices," *Journal of Multivariate Analysis*, vol. 88, no. 2, pp. 365–411, 2004.

[16] S. Pang, D. Morris, and H. Radha, "CLOCs: Camera-LiDAR object candidates fusion for 3D object detection," in *Proceedings of the 2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2020, pp. 10 386–10 393.

[17] Y.-T. Chen, J. Shi, Z. Ye, C. Mertz, D. Ramanan, and S. Kong, "Multimodal object detection via probabilistic ensembling," in *Computer Vision – ECCV 2022*, ser. Lecture Notes in Computer Science, 2022, pp. 139–158.

[18] J. Yoo, D. Lee, I. Chung, D. Kim, and N. Kwak, "What, how, and when should object detectors update in continually changing test domains?" in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024, arXiv:2312.08875.

[19] G. Jocher, A. Chaurasia, and J. Qiu, "Ultralytics YOLOv8," https://github.com/ultralytics/ultralytics, 2023, software, version 8.0.0; this work used the Ultralytics library at version 8.4.90. AGPL-3.0.

[20] C.-Y. Wang, I.-H. Yeh, and H.-Y. M. Liao, "YOLOv9: learning what you want to learn using programmable gradient information," in *Computer Vision – ECCV 2024*, ser. Lecture Notes in Computer Science, 2024, pp. 1–21.

[21] A. Wang, H. Chen, L. Liu, K. Chen, Z. Lin, J. Han, and G. Ding, "YOLOv10: real-time end-to-end object detection," in *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*, 2024.

[22] G. Jocher and J. Qiu, "Ultralytics YOLO11," https://github.com/ultralytics/ultralytics, 2024, software, version 11.0.0. AGPL-3.0.

[23] Y. Tian, Q. Ye, and D. Doermann, "YOLOv12: attention-centric real-time object detectors," in *Advances in Neural Information Processing Systems 38 (NeurIPS 2025)*, 2025, arXiv:2502.12524.